



Does Dodd-Frank affect OTC transaction costs and liquidity? Evidence from real-time CDS trade reports[☆]



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ABSTRACT

This paper examines transaction costs and liquidity in the index CDS market by matching intraday quotes to real-time trade reports made available through the Dodd-Frank reforms. We find that the average relative effective spread is 0.27% of price level or 2.73% of CDS spread. Dodd-Frank does affect transaction costs and liquidity. Liquidity improves after the commencement of public dissemination of OTC derivatives trades. Moreover, cleared trades, trades executed on exchange-like venues, end-user trades, and bespoke trades exhibit lower trading costs, price impact, and price dispersion. These findings improve our understanding of the OTC derivatives market that is undergoing fundamental changes.

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1. Introduction

The cost of trading is a market friction that is of interest to academics, investors, and regulators. Studies have shown that trading costs affect asset prices in a variety of financial markets.² Investors incur trading costs when establishing and liquidating their asset positions, and such costs diminish portfolio value. Regulators are concerned

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² The literature on the relation between trading costs and asset prices covers various markets including the stock market (e.g., Amihud and Mendelson, 1986; Acharya and Pedersen, 2005), the bond market (e.g., Amihud and Mendelson, 1991; Friewald, Jankowitsch, and Subrahmanyam, 2012), and the derivatives market (e.g., Brenner, Eldor, and Hauser, 2001; Deuskar, Gupta, and Subrahmanyam, 2011). See Amihud, Mendelson, and Pedersen (2005) for a survey of this literature.

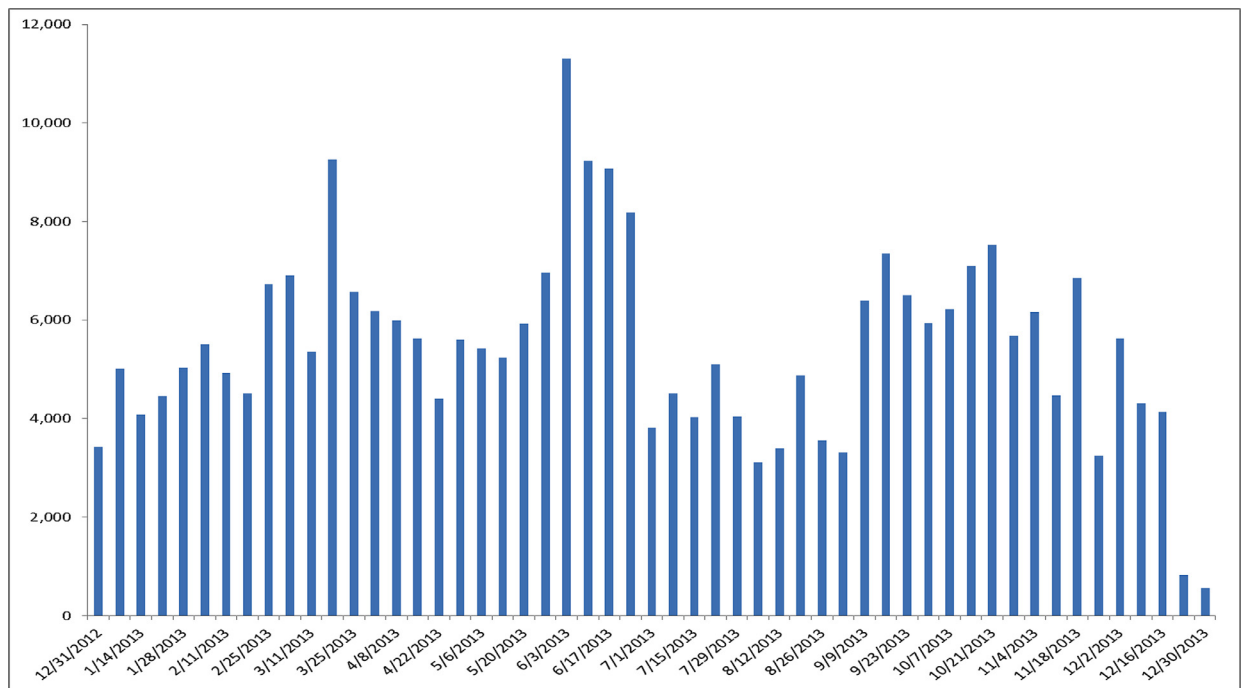


Fig. 1. Weekly credit derivatives trades from DDR reports. This figure presents the weekly number of credit derivatives trades executed between December 31, 2012 and December 31, 2013 and publicly disseminated by DTCC Data Repository (U.S.) LLC (DDR). The sample includes a total of 289,505 trades, of which 267,549 are index CDS trades.

with the functioning of financial markets, and trading costs are important gauges of the ease with which market participants can transact with each other. Furthermore, trading costs and liquidity are linked to systemic risk. A distressed financial institution that conducts a fire sale of securities in an illiquid market with high trading costs can drive significant price declines, which in turn threaten the solvency of other institutions holding the same securities. These compelling reasons notwithstanding, research on the transaction costs and liquidity of the over-the-counter (OTC) derivatives market has been limited, largely due to the lack of publicly available transaction data. In this paper, using real-time trade reports made available by post-financial crisis reforms, we examine the trading costs and liquidity of index credit default swaps (CDSs), an important class of OTC derivatives.³ More importantly, the richness of the disseminated trade reports allows us to analyze how different aspects of regulatory reforms are changing the landscape and thus, the liquidity of the once opaque OTC derivatives market.

As part of the Dodd-Frank Wall Street Reform and Consumer Protection Act, the Commodity Futures Trading Commission (CFTC) implemented mandatory real-time reporting and public dissemination of OTC swap trades on December 31, 2012. Swap transactions have to be reported

to a recordkeeping facility, known as a swap data repository (SDR), which in turn disseminates transaction details (e.g., trade price, trade size, timestamp, and trade characteristics linked to the Dodd-Frank reforms) to the public. We combine publicly disseminated index CDS trades (executed between December 31, 2012 and December 31, 2013) from Depository Trust & Clearing Corporation (DTCC) Data Repository (U.S.) LLC (DDR) with the intraday quote data and end-of-day valuation data from two major CDS data providers, Credit Market Analysis Ltd. (CMA) and Markit Group Ltd. (Markit), to calculate transaction-level relative effective spread and other liquidity measures.⁴ We find substantial trading activity in the index CDS market. On average, 5,462 trades are executed each week (see Fig. 1). The average relative effective spread is about 0.27% of price level or 2.73% of CDS spread, which is in line with the transaction costs reported in other derivatives markets such as the options market.⁵ Moreover, we find that liquidity varies considerably across different families of index CDSs and liquidity is much higher for on-the-run series and the most popular 5-year tenor.

The Dodd-Frank Act fundamentally changes the OTC swap market in several ways. First, it requires real-time

³ The CDS market is important not only because it allows investors to hedge credit risk, but also because it facilitates price discovery of credit risk through informed trading (Acharya and Johnson, 2007). Furthermore, CDSs have been linked to the huge losses sustained by financial institutions in the recent financial crisis and thus have important implications for financial stability.

⁴ Detailed information on real-time trade reporting requirements, descriptions of index CDS contracts, and a description of the data collection and cleaning process are provided in Appendices A, B, and C, respectively.

⁵ See, for example, Goyenko, Ornathanalai, and Tang (2014), Muravyev and Pearson (2014), Augustin, Brenner, Hu, and Subrahmanyam (2015), and Augustin, Brenner, and Subrahmanyam (2015).

reporting and public dissemination of transactions, a fundamental shift in the previously opaque market with little or no post-trade transparency. Second, it mandates central counterparty (CCP) clearing for eligible swaps, which were traditionally traded in a bilateral setting without the disciplining influence of clearing, including the posting of margins. Third, it creates swap execution facilities (SEFs), a new type of trading venue where multiple traders can execute swap transactions by accepting bids and offers posted by other traders. In addition, the Dodd-Frank Act separates nonfinancial entities and small financial institutions that use swaps to hedge or mitigate commercial risk (termed “end-users”) from swap dealers and major swap participants.⁶ The Dodd-Frank Act also distinguishes between standardized and bespoke contracts. Finally, the Dodd-Frank Act separates block trades from non-block trades.

While one of the main objectives of the Dodd-Frank Act is to strengthen liquidity in the OTC swap market, some market participants have expressed concerns that, contrary to the intentions of the Dodd-Frank Act, these reforms might in fact degrade market quality.⁷ In this paper, we first provide evidence of how the Dodd-Frank Act could have affected market quality by studying changes in index CDS liquidity around the CFTC’s implementation of regulatory reforms. We then investigate how trading costs and liquidity are affected by various deal characteristics that came to be defined in the post-Dodd-Frank era.

The first hypothesis we test is whether liquidity improves after the commencement of real-time reporting and public dissemination of OTC swap trades on December 31, 2012. According to the literature, reporting privately negotiated bilateral deals to SDRs represents a major improvement in market transparency in the hitherto opaque CDS market, and public dissemination of key trade information such as price and volume can lower transaction costs and improve liquidity (e.g., Pagano and Roëll, 1996; Naik, Neuberger, and Viswanathan, 1999; Bessembinder and Maxwell, 2008; Acharya, Engle, Figlewski, Lynch, and Subrahmanyam, 2009). Using data from the pre-Dodd-Frank period, we find that the initiation of real-time reporting and public dissemination of OTC swap trades are indeed associated with transaction cost reductions and liquidity improvements for index CDS contracts. These results are obtained after accounting for liquidity effects due to index segments, on-the-run status, and tenor. For example, the relative quoted spread of the 5-year on-the-run CDX NA IG index (the most popular index CDS contract) decreases from 0.64% in the pre-Dodd-Frank period to 0.56% after the start of trade reporting and dissemination, and the decline is statistically significant at the 1% level. Similar findings are also obtained for other liquidity measures

and for other index CDS contracts. (See Appendix B for descriptions of various index CDS contracts.)

The second hypothesis we examine is how various deal characteristics that came to be defined in the post-Dodd-Frank era affect transaction costs and liquidity. In addition to transaction price and size, the disseminated trade reports provide a rich set of trade characteristics that capture different aspects of regulatory reform.⁸ We use these deal characteristics to evaluate a number of hypotheses relating different aspects of Dodd-Frank regulations to liquidity. Following Friewald, Jankowitsch, and Subrahmanyam (2014), we focus on conceptually sound, transaction-based liquidity measures that have been proposed for OTC markets: the relative effective spread, the Amihud measure (Amihud, 2002), the Jankowitsch, Nashikkar, and Subrahmanyam price dispersion measure (Jankowitsch, Nashikkar, and Subrahmanyam, 2011), and the Roll measure (Roll, 1984). Our empirical strategy exploits cross-sectional variations in liquidity across trades exhibiting different Dodd-Frank deal characteristics. Specifically, we sort the reported trades into mutually exclusive buckets distinguished by different sets of deal characteristics and calculate daily liquidity measures for each bucket. We then conduct multivariate analysis to examine the effects of these deal characteristics on the liquidity measures.

The first characteristic we examine is central clearing, which is an effective way to reduce counterparty risk and increase post-trade transparency according to the existing literature (Acharya, Engle, Figlewski, Lynch, and Subrahmanyam, 2009; Acharya and Bisin, 2014). We find that centrally cleared trades are associated with higher liquidity than noncleared trades. Specifically, cleared trades have lower transaction costs, price impact, and price dispersion than noncleared trades, and the differences are statistically significant. These results suggest that the reduced counterparty risk and increased post-trade transparency associated with central clearing have beneficial effects on liquidity.

Besides central clearing, we find that aspects of the regulatory reforms that have not been previously studied also have significant effects on trading costs and liquidity. For example, trades executed on SEFs have lower transaction costs, price impact, and price dispersion than non-SEF trades, which is consistent with the prediction that increased pre-trade transparency improves liquidity (e.g., Flood, Huisman, Koedijk, and Mahieu, 1999; Hendershott and Madhavan, 2015). End-user trades exhibit higher liquidity and lower trading costs, which supports the notion that qualifying for end-user status validates nonfinancial and small financial hedgers as uninformed traders. We also find lower transaction costs, price impact, and price dispersion for bespoke trades. We obtain these results after controlling for liquidity effects associated with trading activities (trading volume, number of trades, trade interval, number of dealers, and number of quotes), on-the-run status, index CDS tenor, index family, and product characteristics (maturity and coupon).

⁶ See CFTC’s fact sheet, “Final Rule on End-User Exception to the Clearing Requirement for Swaps” for details (http://www.cftc.gov/ucm/groups/public/@newsroom/documents/file/eue_factsheet_final.pdf).

⁷ See, for example, Peter Madigan, “US end-users are losers in swaps liquidity split”, *Risk Magazine*, April 28, 2014; Matt Cameron, “Pension funds snub swaps over margin demand”, *Risk Magazine*, April 2, 2014; Pam Brown, “Derivatives regulatory concerns of end-users. What should treasurers do?” *Financial Executive*, May 1, 2012; Peter Madigan, “SEFs threaten dealer-client relationship”, *Risk Magazine*, April 28, 2011.

⁸ See Appendix A for a list of data fields in the disseminated trade reports that are related to the Dodd-Frank Act.

Next, to take advantage of the high granularity of the trade-by-trade information made available by swap trade reporting, we extend our investigation to examine whether the various Dodd-Frank reforms have distinct, incremental effects on trading cost at the high-frequency, transaction level. For this purpose, we leverage the matching of trade reports to intraday quotes, and the fact that Dodd-Frank deal characteristics are specifically defined for each individual trade. We focus on trade-specific relative effective spread as a measure of round-trip trading cost because in dealer-intermediated derivatives markets, transactions can occur within or outside quoted bid and offer prices due to negotiations between customers and dealers. These negotiations are driven by price-affecting factors, such as adverse selection, inventory cost, search cost, and dealer market power, which are in turn affected by the Dodd-Frank reforms.

We estimate transaction-level regressions of the relative effective spread on the Dodd-Frank deal characteristics and a variety of control variables. Our regression results show that clearing, SEF trading, end-user status, and contract customization have separate, incremental effects on trading costs. For example, in the CDS spread-quoted sample, the relative effective spread is lower by 0.45% for cleared trades compared to noncleared trades. Trades executed on SEFs incur relative effective spreads that are lower by 0.47%. Trading cost reductions for end-user trades and bespoke trades are 0.69% and 0.81%, respectively. In addition, relative effective spreads decline with trade size and are narrower for on-the-run trades and 5-year trades. Our findings continue to hold when we repeat the analysis for the four most actively traded index CDSs (North American index CDSs: CDX NA IG and CDX NA HY; European index CDSs: iTraxx Europe and iTraxx Crossover).

The CFTC implemented mandatory clearing for these four index CDSs in three phases for different groups of market participants (see Fig. 2, Panels A and B). This phased introduction provides a natural experiment to test for the liquidity effects of various Dodd-Frank deal characteristics while avoiding contamination by endogeneity issues. In our last analysis, we estimate an augmented transaction-level regression that includes indicator variables for the three phases and interactions between the indicator variables and deal characteristics associated with mandatory clearing. Coefficients on the interaction terms indicate that the liquidity enhancing effects of clearing, end-user status, and contract customization grow as the coverage of mandatory clearing expands. These results strengthen our finding that the Dodd-Frank deal characteristics are related to CDS transaction costs and liquidity.

The rest of the paper proceeds as follows. Section 2 reviews various strands of research related to our study and Section 3 develops our hypotheses. In Section 4, we describe the composition of the CDS trade reports and discuss how they are linked to intraday and end-of-day data. We also identify data issues in the trade reports and discuss how to address them. In Section 5, we define trading costs and other liquidity measures and report their summary statistics. In Section 6, we present the empirical results. Various robustness-checks are discussed in Section 7 and concluding remarks are in Section 8.

2. Literature review

In this section, we discuss related research and highlight our contributions to the literature.⁹ Our study contributes to the literature in a number of ways. First, the literature on credit spread often implicitly assumes that the CDS spread is a pure measure of credit risk. However, research suggests that the CDS market is not perfectly liquid (e.g., Bongaerts, De Jong and Driessen, 2011; Tang and Yan, 2014, 2015). Nevertheless, quantifying liquidity and trading costs in the opaque over-the-counter CDS market has been challenging to researchers because of the lack of market-wide and publicly available transaction data. Furthermore, the effects of market transparency and public trade disclosure on liquidity and transaction costs are not well understood for this market. By analyzing various aspects of the roll-out of the Dodd-Frank reforms, including publicly available intraday trade reports, we show that liquidity improves after the start of OTC swap trade dissemination and that liquidity is related to deal characteristics defined in the post-Dodd-Frank era. In addition, we uncover data issues in the publicly disseminated trade reports and offer systematic ways to address those issues. Given the research potential of these data, we believe our findings will be useful to readers with an interest in credit markets.

Second, our paper adds to the literature on the post-financial crisis market reforms. Studies such as those undertaken by Acharya, Engle, Figlewski, Lynch, and Subrahmanyam (2009), Pirrong (2009), Duffie and Zhu (2011), Acharya and Bisin (2014), and Cont and Kokholm (2014) analyze the implications of derivatives market reforms from a theoretical perspective. Compared to the theoretical literature, the empirical evidence on the post-crisis reforms is limited and focuses exclusively on the effects of central clearing (e.g., Loon and Zhong, 2014; Duffie, Scheicher, and Vuillemeys, 2015). In contrast, using newly available information from the real-time trade reports for index CDSs, this paper examines how liquidity is related to different aspects of the Dodd-Frank reforms, including clearing, SEF, end-user status, bespoke contracts, and block trades.

Third, we contribute to the literature that examines trading and liquidity in the CDS market. Chen, Fleming, Jackson, Li, and Sarkar (2011) analyze daily trades in index and single-name CDSs over a three-month period and show that trading is concentrated in standardized contracts. Shachar (2012) and Gündüz, Nasev, and Trapp (2013) employ daily order flow data to study the price impact of single-name CDSs. Oehmke and Zawadowski (2015) use weekly positions and volume data for single-name CDSs and find that hedging and speculation are determinants of trading in the single-name CDS market. In contrast to these studies, we examine the incremental impact of various Dodd-Frank reforms on trading costs and liquidity in the index CDS market.

Finally, our paper is also related to the recent strand of literature on liquidity measurement in fixed income

⁹ See Augustin, Subrahmanyam, Tang, and Wang (2014) for an excellent survey of research on CDS.

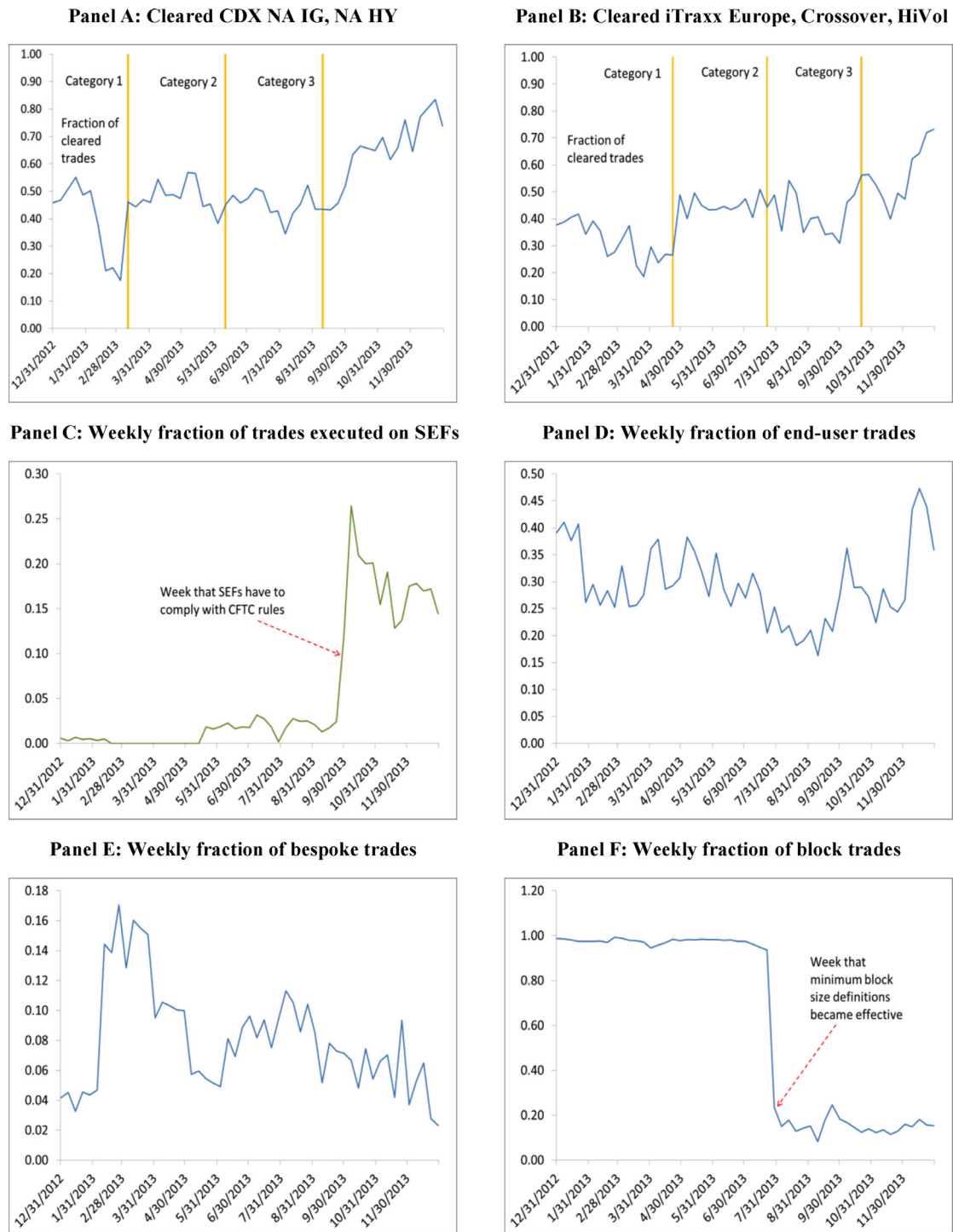


Fig. 2. Weekly fraction of cleared trades, SEF, end-user, bespoke, and block trades. Panel A (B) presents plots of the weekly fraction of cleared trades in CDX indexes (iTraxx indexes) subject to mandatory clearing. Vertical lines mark out the commencement of mandatory clearing for different categories of market participants. The specific tenor and series covered by the clearing requirement are provided in [Appendix A.2.1](#). Panels C, D, E, and F present, respectively, the weekly fraction of trades executed on SEFs, end user trades, bespoke trades, and block trades. SEFs were required to comply with CFTC rules on registration and operation by October 2, 2013. CFTC's minimum block size definitions became effective on July 30, 2013.

markets.¹⁰ In particular, Friewald, Jankowitsch, and Subrahmanyam (2014) study liquidity in the fixed-income securitized product market using various measures based on different levels of detail in transaction data. We complement their study by quantifying liquidity in the CDS market using measures with different input requirements in terms of trade and quote data. Using transaction data for the corporate and municipal bond markets, researchers find that trading costs decline with trade size (Schultz, 2001; Bessembinder, Maxwell, and Venkataraman, 2006; Harris and Piwowar, 2006; Edwards, Harris, and Piwowar, 2007; Goldstein, Hotchkiss, and Sirri, 2007; Green, Hollifield, and Schürhoff, 2007). Consistent with these studies, we find that trading costs also decline with trade size in the CDS market.

3. Hypothesis development

In this section, we build on insights from the existing literature and formulate hypotheses to be tested in our study. We lay the groundwork for hypothesis development by discussing salient features of the index CDS market, which is the focus of our empirical investigation. Specifically, we state the key difference between a single-name CDS and an index CDS, highlight the importance of index CDS contracts in the global CDS market, describe major types of market participants, discuss the balance between standardization and customization, and establish the representativeness of the main data set used in the paper.

Whereas a single-name CDS contract provides default protection on an individual reference entity, an index CDS contract offers default protection on a group of reference entities representing a well-defined segment of the credit market (Appendix B describes the most popular index CDSs). The index CDS market constitutes a significant share of the overall CDS market. At the end of June 2014, the global CDS market had an aggregate notional value of \$19.4 trillion, of which \$7.9 trillion (41%) was in index CDS contracts (Bank for International Settlements, 2014). Thus, activity in the index CDS sector can provide useful insights into the workings of the overall CDS market.

Under the Dodd-Frank regime, key participants in the index CDS markets are swap dealers, major swap participants, and customers. Besides trading index CDSs on their own behalf, swap dealers offer liquidity to their customers. Major swap participants are non-dealer financial institutions that trade actively in the swap market. Customers in the index CDS market mainly consist of financial institutions such as non-dealer banks, insurance companies, hedge funds, and other institutional investors that use index CDSs for hedging or speculation (Markit, 2014). In addition, nonfinancial entities, such as Microsoft, can also use

credit derivatives such as index CDS to manage their risk exposure.¹¹

Under the Dodd-Frank Act, nonfinancial entities and small financial institutions (banks, savings associations, farm credit system institutions, and credit unions with total assets of \$10 billion or less) that solely use swaps to hedge commercial risks can apply to the CFTC to qualify as end-users.¹² Index CDSs have become popular hedging instruments among end-users because it is often less costly to use an index CDS to mitigate the credit risk of a portfolio than to trade the single-name CDSs of the portfolio's underlying reference entities (Markit, 2014). End-users are exempted from certain regulations (such as mandatory clearing).¹³ While most of the traded index CDS contracts are standardized, some contracts are customized with unique terms and conditions (such as payment structures and payment dates) to match the risk profiles of the transacting parties. Regulators across the globe have been trying to strike a balance between fostering greater product standardization and preserving access to bespoke contracts in OTC derivatives markets.¹⁴ As a result, both standardized and bespoke index CDS contracts are still being traded under the new regulatory regime.

High-frequency data on index CDS trading activity became publicly available for the first time with the commencement of the Dodd-Frank mandated real-time trade reporting. During the first year of this new reporting regime, three SDRs were in operation: DDR, Intercontinental Exchange Inc., SDR (ICE), and Chicago Mercantile Exchange Inc. SDR (CME).¹⁵ Among these three SDRs, DDR covers about 99.5% (387,696) of all publicly disseminated credit derivatives trades executed between December 31, 2012 and December 31, 2013 and ICE covers the remaining

¹¹ According to its 10-K reports, Microsoft has been using CDSs for risk management since around 2010. For instance, in its 10-K for the fiscal year ended June 30, 2011, Microsoft states that, "We use credit default swaps as they are a low cost method of managing exposure to individual credit risks or groups of credit risks. As of June 30, 2011, the total notional amounts of credit contracts purchased and sold were \$532 million and \$277 million, respectively. As of June 30, 2010, the total notional amounts of credit contracts purchased and sold were \$371 million and \$199 million, respectively." Microsoft is a member of the Coalition of Derivatives End-Users, an organization that represents the views of companies that use derivatives primarily to manage risks associated with their businesses (<http://coalitionforderivativesendusers.com/home/>).

¹² Specifically, CFTC Regulation Section 39.6(d) exempts banks, savings associations, farm credit system institutions, and credit unions with total assets of \$10 billion or less from the definition of "financial entity," making such "small financial institutions" eligible for end-user status.

¹³ Investors with end-user status still have the option of voluntarily clearing their trades through CCPs. For example, after mandatory clearing was fully phased in, we find that 59% of end-user trades were centrally cleared.

¹⁴ See, for example, the discussion by the Committee of European Securities Regulators (CESR), "...firms should be able to retain the flexibility to customize...given the role that OTC derivatives, and in particular bespoke products, play in meeting hedging needs. Nevertheless, CESR is of the view that greater standardisation of OTC derivatives contracts can deliver efficiency benefits to the market." (CESR consultation paper, "Standardisation and exchange trading of OTC derivatives," July 19, 2010.)

¹⁵ The fourth SDR, operated by Bloomberg L.P., did not begin its operations and public dissemination of credit derivatives trade reports until May 12, 2014, which is more than 16 months after the commencement of Dodd-Frank mandated public dissemination of OTC derivatives trades.

¹⁰ Studies that analyze various liquidity measures in the corporate bond market include Mahanti, Nashikkar, Subrahmanyam, Chacko, and Mallik (2008), Bao, Pan, and Wang (2011), Jankowitsch, Nashikkar, and Subrahmanyam (2011), Lin, Wang, and Wu (2011), Nashikkar, Subrahmanyam, and Mahanti (2011), Feldhütter (2012), Dick-Nielsen, Feldhütter, and Lando (2012), and Pelizzon, Subrahmanyam, Tomio, and Uno (2013, 2015)).

0.5% (2,110), while CME does not report any credit derivatives trades during this time frame. Therefore, our DDR sample covers virtually all index CDS transactions publicly disseminated in the first year of the Dodd-Frank regulatory regime, which allows us to test the effects of the Dodd-Frank regulatory reforms on index CDS transaction costs and liquidity.

Hypothesis 1. Index CDS liquidity improves after the start of real-time reporting and public dissemination of index CDS trades.

In the pre-Dodd-Frank period, OTC swaps were executed in relatively opaque markets with little or no public disclosure of trade information. Under the aegis of the Dodd-Frank Act, real-time reporting and public dissemination of privately negotiated OTC swap transactions via SDRs represents a major improvement in post-trade transparency (Acharya, Engle, Figlewski, Lynch, and Subrahmanyam, 2009). The literature on post-trade transparency suggests that public dissemination of key trade information such as price and volume can improve liquidity by lowering the threat of informed trading to dealers, improving dealer inventory risk sharing, and providing customers with better pricing information (e.g., Pagano and Roëll, 1996; Naik, Neuberger, and Viswanathan, 1999; Bessembinder and Maxwell, 2008). Therefore, we expect higher liquidity and lower trading cost in the index CDS market after trades are promptly reported and disseminated to the public. One counter-argument to the prediction is that real-time public dissemination of swap trades exposes dealers to predatory trading and increases their costs of off-setting customer trades.¹⁶ As a result, dealers curtail liquidity provision or pass on the increased costs of hedging exposures to customers (Brunnermeier and Pedersen, 2005). Another counter-argument relates to the effect of post-trade transparency on interdealer competition. Trade dissemination provides dealers with information previously gleaned from accommodating customer trades. Consequently, dealers have weaker incentives to compete for order flow, resulting in wider bid-ask spreads (Bloomfield and O'Hara, 1999).

Hypothesis 2.1. Transaction costs are lower and liquidity is higher for centrally cleared trades.

When a swap contract is centrally cleared, the CCP guarantees contractual fulfillment, imposes margin requirements, and disseminates clearing and settlement statistics. According to the existing literature, central clearing can reduce or eliminate counterparty risk, provided that there is effective risk management by the

CCP (Acharya, Engle, Figlewski, Lynch, and Subrahmanyam, 2009; Acharya and Bisin, 2014; Loon and Zhong, 2014). We expect higher liquidity for centrally cleared trades for a number of reasons. If margin requirements imposed by the CCP reduce the returns to informed trading in cleared index CDSs, informed traders will move away from such swaps to other credit-sensitive instruments. Consequently, the diminished threat of informed trading increases liquidity in centrally cleared index CDSs (John, Koticha, Narayanan, and Subrahmanyam, 2003). In addition, central clearing equalizes counterparty risk across market participants, allowing smaller dealer banks to compete with larger dealer banks. Increased competition among dealers for order flow reduces bid-ask spreads and improves liquidity. The counter-argument is that central clearing could reduce netting efficiency in some cases and thus require dealers to use more collateral for cleared trades (Duffie and Zhu, 2011). Greater collateral requirements increase dealers' market-making costs and thus reduce liquidity provision.

Hypothesis 2.2. Transaction costs are lower and liquidity is higher for trades executed on SEFs.

An SEF is a new type of exchange-like trading venue created by the Dodd-Frank Act, which represents an increase in pre-trade transparency and opens up the CDS market to a much broader range of participants (Acharya, Engle, Figlewski, Lynch, and Subrahmanyam, 2009).¹⁷ Increased pre-trade transparency can lower bid-ask spread by reducing search costs and increasing quote competition among traders (Flood, Huisman, Koedijk, and Mahieu, 1999; Hendershott and Madhavan, 2015). Therefore, we expect transaction costs are lower and liquidity is higher for trades executed on SEFs. The counter-argument is that greater pre-trade transparency aggravates front-running or picking-off risk, which can increase trading costs and reduce market liquidity (Harris, 1996; Bessembinder, Panayides, and Venkataraman, 2009).

Hypothesis 2.3. Transaction costs are lower and liquidity is higher for trades with end-users.

As mentioned earlier, to qualify as an end-user under the CFTC regulations, an entity must be a nonfinancial entity or small financial institution that uses swaps to hedge its commercial or business risk. In other words, qualifying for end-user status validates the nonfinancial or small financial hedgers as uninformed traders. Therefore, we expect trades executed by end-users to be associated with lower transaction costs and higher liquidity, because uninformed traders pose a lower threat of adverse selection risk to CDS dealers (e.g., Glosten and Milgrom, 1985). The counter-argument is that end-users could have less bargaining power with dealers who prefer to offer better price improvements to regular customers (Bernhardt, Dvoracek, Hughson, and Werner, 2005). In this case, we

¹⁶ As the Dodd-Frank Act was being finalized by lawmakers, dealer banks voiced concerns over the negative effect of real-time public trade dissemination on their ability to hedge. "...For their part, banks have argued real-time reporting of derivatives trades to the public would damage liquidity, as it would make it more difficult to hedge exposures. If dealers were required to disclose trades immediately, before they have had a chance to square their positions, there is a risk the market could move against them, they say. The only entity to benefit would be a third party that gets to glimpse a competitor's trading strategy, they argue." (Matt Cameron, "Risk Senate Ag committee to take OTC derivatives transparency to new levels," *Risk Magazine*, April 15, 2010).

¹⁷ SEFs offer two modes of execution: (a) a limit order book where firm bids and offers are visible to participants, and (b) a request-for-quote (RFQ) system that allows a participant to solicit firm quotes from multiple market participants before deciding to trade (CFTC Rule Part 37).

expect end-user trades to have higher transaction costs and lower liquidity.

Hypothesis 2.4. Transaction costs are lower and liquidity is higher for bespoke trades.

As discussed above, securities regulators stress the need to preserve bespoke contracts under the post-financial crisis regulatory regime because such derivatives are often used by hedgers to mitigate the risks associated with their unique risk profiles (Committee of European Securities Regulators, 2010).¹⁸ We expect bespoke contracts to incur lower transaction costs due to reduced adverse selection risk if these contracts are mostly being used for hedging. The counter-argument is that more sophisticated agents, such as hedge funds, with their potential advantages in pricing complex and innovative contracts, can also trade bespoke contracts. The presence of such sophisticated agents can raise the threat of adverse selection to CDS dealers. Moreover, liquidity could be worse for bespoke contracts because their uniqueness increases inventory holding costs for CDS dealers (Jankowitsch, Nashikkar, and Subrahmanyam, 2011). Because of these counter-arguments, bespoke contracts could have higher transaction costs and poorer liquidity.

Hypothesis 2.5. Transaction costs are lower and liquidity is higher for block trades.

CFTC allows block trades to be publicly disseminated with a delay in contrast to non-block trades, which must be disseminated as soon as technologically practicable after execution. The rationale for reporting delays is to allow swap dealers sufficient time to conduct off-setting trades without bearing undue costs. If blocks are traded for hedging purposes, then we expect lower transaction costs and higher liquidity for block trades due to a lower risk of adverse selection (Seppi, 1990; Madhavan and Cheng, 1997). The counter-argument is that informed traders may prefer to submit large orders to maximize the benefit of their private information. In this case, block trades pose a greater threat of informed trading, resulting in higher trading costs and poorer liquidity (Easley and O'Hara, 1987).

4. Data and sample

4.1. Real-time trade reports

We obtain 387,696 reports of OTC credit derivatives trades executed between December 31, 2012 and December 31, 2013 and publicly disseminated by DDR. Each DDR report has a numeric identifier (DISSEMINATION_ID), execution timestamp, price, trade size (expressed as the notional amount of the CDS contract), currency denomination, and other contract characteristics.¹⁹ A report can indicate a new trade, cancel a previous report, or correct a

previous report. To cancel a previous report, the reporting entity files a cancellation report that identifies the original report to be canceled. To correct a report, the reporting entity must not only cancel the original erroneous report but also file a new report with the correct information (correction report). Like the cancellation report, the correction report also indicates the original report's identifier. We use this reporting protocol to perform an initial cleaning of the data. First, we remove duplicate reports (i.e., reports with the same identifier). Next, we remove canceled reports and cancellation reports. Finally, we remove duplicate correction reports and correction reports that cannot be linked to the original reports due to a missing identifier for the original transaction. The cleaning process reduces the sample from 387,696 to 289,505 reports of which 272,055 are new reports and 17,450 are correction reports.²⁰

Table 1 describes the composition of the trade reports. Panel A presents the distribution of reports across different types of transactions. There are 198,416 new trades, which constitute 69% of the sample. The remaining reports cover transactions related to lifecycle events such as partial termination, novation, and amendments. Panel B shows that the reported transactions encompass a wide range of credit derivatives, with the lion's share taken up by untranch index CDS (267,549 trades; 92% of the sample). Furthermore, among this category, the bulk of the transactions are for index CDSs referencing corporate bond issuers. Panel C reports the level of collateralization in reported trades. About 44% of the trades are backed by some form of collateralization. Specifically, 24% require counterparties to post initial and variation margin (fully collateralized), 19% require counterparties to provide variation margin only, and 1% of the sample requires only one of the counterparties to post margin. About one-quarter of the transactions are uncollateralized. Panel D shows the distribution of currency denominations. Transactions are concentrated in US dollar-denominated contracts (69% of the sample). Euro-denominated and Yen-denominated contracts make up 30% and 1% of the sample, respectively.

4.2. Trade characteristics related to the Dodd-Frank reforms

The DDR reports provide a rich set of trade characteristics that capture different aspects of regulatory reform. (Appendix A provides further information on the Dodd-Frank reforms and maps them to the DDR data fields.) Table 2 shows the frequency distribution of each Dodd-Frank deal characteristic among credit derivatives trades.

First, during our sample period, about 25% of the trades are centrally cleared either voluntarily or in compliance with the regulatory mandate. The CFTC implemented mandatory clearing in 2013 for a set of actively traded index CDSs (CDX NA IG, CDX NA HY, iTraxx Europe, iTraxx Europe Crossover, and iTraxx Europe HiVol) in three phases. (See Appendix A.2.1 for details.) Fig. 2 Panel

¹⁸ This point is echoed by practitioners. See, for example, Christian Weistroffer, "Credit default swap: Heading towards a more stable system," Deutsche Bank Research, December 21, 2009.

¹⁹ DDR's Real Time Dissemination Dashboard User Guide provides a complete list of data fields in publicly disseminated trade reports (<https://rtdata.dtcc.com/gtr/dashboard.do>).

²⁰ The CFTC also requires reporting of swaps entered into prior to December 31, 2012 (historical swaps). In our sample, we find 5,622 historical swap trade reports after our initial cleaning process. Given the relatively small number of these reports (1.9% of the number of real-time trade reports), we do not include them in our analysis.

Table 1

Composition of DDR trade reports.

This table reports the distribution of OTC credit derivatives trades across transaction types, types of credit derivatives, collateralization level, and currency denomination. We obtain trades executed between December 31, 2012 and December 31, 2013 and publicly disseminated by DTCC Data Repository (U.S.) LLC (DDR).

<i>Panel A: Transaction type</i>		
	<i>N</i> (total = 289,505)	Pct of total (%)
Trade	198,416	68.54
Partial termination	51,944	17.94
Novation	21,149	7.31
Amendment	17,975	6.21
Exercise	20	0.01
Increase	1	0.00
<i>Panel B: Types of credit derivatives</i>		
	<i>N</i> (total = 289,505)	Pct of total (%)
Index CDS		
Corporate bond issuers	194,588	67.21
Municipal bond issuers	803	0.28
Sovereign bond issuers	8,883	3.07
Corporate loan issuers	179	0.06
Structured finance	63,096	21.79
Swaptions	11,329	3.91
Index tranche	7,936	2.74
Exotic	1,584	0.55
Total return swap	661	0.23
Corporate	363	0.13
Sovereign	83	0.03
<i>Panel C: Collateralization</i>		
	<i>N</i> (total = 289,505)	Pct of total (%)
Fully collateralized	70,592	24.38
One-way collateralized	3,459	1.19
Partially collateralized	53,828	18.59
Uncollateralized	69,172	23.89
Not reported	92,454	31.94
<i>Panel D: Currency</i>		
	<i>N</i> (total = 289,505)	Pct of total (%)
USD	199,440	68.89
EUR	86,293	29.81
JPY	3,279	1.13
Others	493	0.17

A presents the weekly fraction of cleared trades in the CDX indexes subject to mandatory clearing along with vertical bars marking the start of each phase. Prior to the commencement of mandatory clearing, some trades in these indexes were already voluntarily cleared through CCPs. In the week of the first compliance date, the fraction of cleared CDX trades jumps from 17% to 46% and then hovers around 50% until the last category of swap users began mandatory clearing. Thereafter, the fraction of cleared trades gradually rises to end at a high of 83% shortly before the end of 2013. The weekly plot for iTraxx indexes (Fig. 2, Panel B) exhibits a similar pattern, with the fraction of cleared trades hitting a high of 73% in the last week of 2013.

Second, over our entire sample period, we find that only 14,588 credit derivatives trades (5% of the sample) are executed on SEFs while the rest are executed in the traditional bilateral trading environment. However, these ba-

sic statistics mask interesting time series patterns in SEF usage. SEFs were required to comply with CFTC rules on registration and operation by October 2, 2013 (start date). When we examine the period from October 2 to December 31, 2013, we find the proportion of SEF trades increases to 18%. This pattern is further illustrated in Panel C of Fig. 2.

Third, in the DDR data, we identify 84,080 end-user trades. Of the remaining 205,425 trades, 36,387 are explicitly listed as non-end-user trades (i.e., trades involving no end-users), while 169,038 trades have no information on end-user status, which we also classify as non-end-user. Fig. 2 Panel D shows that the fraction of end-user trades generally hovers around 30% during the sample period. In addition, in our data set, 8% of the trades are customized while the remaining 92% are trades in standardized contracts. Fig. 2 Panel E shows that the weekly fraction of bespoke trades varies between 2% and 17% during 2013.

Finally, 66% of the sample is reported as block trades, while the balance is reported as non-block trades. Part of the explanation for this counterintuitive finding is that the CFTC's definitions for block sizes only became effective on July 30, 2013. Prior to this date, it is possible that counterparties used their own notion of a block trade when submitting reports, resulting in a preponderance of self-declared block trades. Using only reports filed on or after July 30, we find that only 15% of the trades are classified as block trades. We illustrate these findings graphically in Fig. 2 Panel F. Prior to July 30, the fraction of block trades stays close to 100% but plunges to approximately 25% in the last week of July and never exceeds that level for the rest of the year. Our finding indicates that the block trade data field in the DDR report is not informative before July 30, 2013, and researchers should use this variable with care.

4.3. Matching trade reports to intraday quotes and end-of-day valuations

A key objective of our research is to study liquidity patterns in the CDS market using actual prices and trade sizes obtained from the DDR reports. Given the multifaceted nature of liquidity (e.g., effective spread, price impact, price dispersion, and depth), we construct a variety of liquidity measures, many of which require intraday quote data and end-of-day valuations. We therefore match the DDR trade reports to intraday quote data from CMA and end-of-day valuation data from Markit. The Markit and CMA data cover a broad cross-section of the index CDS market including indexes on corporate bond issuers (e.g., CDX NA IG, iTraxx Europe), municipal bond issuers (MCDX), and sovereign bond issuers (e.g., CDX EM). By linking the DDR reports to the Markit and CMA data, we construct a sample whose broad coverage allows us to conduct a comprehensive study of liquidity in the index CDS market.

The DDR trade reports contain a number of data issues, which have been acknowledged by the CFTC.²¹ The

²¹ See CFTC Commissioner O'Malia's keynote address to the Securities Industry and Financial Markets Association (SIFMA) Compliance and Legal Society Annual Seminar, "CFTC's Implementation of Dodd-Frank—Grading Agency Transparency," March 19, 2013. <http://www.cftc.gov/PressRoom/SpeechesTestimony/opamalia-22>.

Table 2

Characteristics related to regulatory reforms.

This table reports the distribution of 289,505 OTC credit derivatives trades across characteristics related to the Dodd-Frank regulatory reforms. We obtain trades executed between December 31, 2012 and December 31, 2013 and publicly disseminated by DDR. Characteristics related to the Dodd-Frank reforms are extracted from DDR trade reports. Cleared indicates that the trade is cleared through a central counterparty. SEF means that the trade is executed on a swap execution facility or designated contract market. A swap execution facility is a trading venue in which multiple traders can execute swap transactions by accepting bids and offers posted by other traders (Dodd-Frank Act, Section 721). We report the distribution of SEF trades for the entire sample period and for the subperiod starting on October 2, 2013, when swap execution facilities have to comply with CFTC rules on registration and operation. End-user indicates that at least one party to the trade is a nonfinancial entity that engages in swap transactions to hedge business risks (Commodity Exchange Act Section 2(h)(7) and CFTC Regulation 39.6). Bespoke means that the trade is in a customized CDS contract, which contains price-affecting terms that are not required to be disclosed in the DDR trade report. Block means that the trade is a block trade. We report the distribution of block trades for the entire sample period and for the subperiod starting on July 30, 2013, when CFTC's block size definitions came into effect.

	Yes		No	
	Number	Pct of total (%)	Number	Pct of total (%)
Cleared	73,685	25.45	215,820	74.55
SEF	14,588	5.04	274,917	94.96
SEF: Beginning October 2, 2013	12,237	18.47	54,004	81.53
End-user	84,080	29.04	205,425	70.96
Bespoke	23,745	8.20	265,760	91.80
Block	190,999	65.97	98,506	34.03
Block: Beginning July 30, 2013	16,594	15.01	93,970	84.99

problems we uncover here include incomplete price information, erroneously reported prices, inconsistency in trade size reporting, and anomalous price deviations from market consensus valuation. These data issues notwithstanding, the trade reports offer crucial data (e.g., price and trade size) during a time period when significant regulatory reforms were being implemented in the OTC derivatives market. As such, the real-time reports are indispensable for the critical task of evaluating the impact of regulatory reforms. Hence, we bring to bear all available data resources (from CMA and Markit) to address the data issues and construct a final sample that allows us to examine the effects of the Dodd-Frank reforms at this critical juncture. (Appendix C describes how we match the DDR reports to CMA and Markit and also provides detailed procedures for addressing the abovementioned data issues in the trade reports.)

After linking the DDR trade reports to CMA and Markit and cleaning the linked reports, we arrive at the final sample of 165,919 trades executed between December 31, 2012 and December 31, 2013. This sample consists of 117,377 trades in spread-quoted contracts and 48,542 trades in price-quoted contracts. All subsequent analyses are based on this sample, unless otherwise stated.

5. Liquidity measures

In this section, we describe the calculation of different measures of liquidity for index CDSs using the newly available trade reports in the post-Dodd-Frank era.²² In addition, we also describe the calculation of more basic measures of liquidity using data from the pre-Dodd-Frank period.

²² For the purpose of calculating liquidity measures, we characterize an index CDS contract as a specific combination of index, series, and tenor. To ensure a consistent interpretation of trading volume, we convert Euro- and Yen-denominated trade sizes to their US dollar equivalents using daily exchange rates from the Federal Reserve Bank of New York.

5.1. Measures of liquidity in the post-Dodd-Frank era

5.1.1. Transaction cost measures

The effective spread for the k th trade in a CDS contract on day t , ES , is $ES = 2 \times |p_k - m_k|$, where p_k is the transaction price of the k th trade, and m_k is the prevailing quote midpoint at the time of the trade.²³ We define the prevailing quote for a trade as the most recent quote before the trade, provided the quote occurs on the same day as the trade. The relative effective spread for the k th trade in a CDS contract on day t , RES , is

$$RES = (ES/m_k) \times 100. \quad (1)$$

For transaction-level analyses, we use ES and RES calculated for each individual trade. For daily analyses, we calculate the daily weighted averages of transaction-level ES and RES , where weights are individual trade sizes.

Quoted spread for the k th trade in a CDS contract on day t , QS , is $a_k - b_k$, where a_k (b_k) is the prevailing ask (bid) quote at the time of the trade. Relative quoted spread for the k th trade in a CDS contract on day t , RQS , is

$$RQS = (QS/m_k) \times 100. \quad (2)$$

At the daily level, QS and RQS are the equally weighted averages of transaction-level QS and RQS , respectively.

As a robustness check, we also compute Roll's (1984) estimate of effective spread. For a CDS contract on day t , the Roll (1984) estimate, $Roll$, is $\sqrt{-\text{cov}(\Delta p_k, \Delta p_{k-1})}$, where Δp_k and Δp_{k-1} are a pair of consecutive transaction price changes on day t with Δp_k (Δp_{k-1}) defined as the change in transaction prices from trade $k-1$ to k ($k-2$ to $k-1$).²⁴ To estimate the serial covariance in price

²³ Both the timestamps of CMA quotes and DDR trade reports are accurate to the second.

²⁴ Each DDR report has a unique numeric identifier and an execution timestamp in Coordinated Universal Time (UTC) with two digits for each of the hour, minute, and second. Therefore, our data satisfy the requirements of the various liquidity measures, including the accurate sequencing of trades as required in the derivation of the Roll measure.

changes, we require at least three consecutive transaction price changes on day t . Following [Friedwald, Jankowitsch, and Subrahmanyam \(2012\)](#), we set positive serial covariances to zero when calculating *Roll*.

5.1.2. Price impact and price dispersion measures

We use [Amihud's \(2002\)](#) illiquidity ratio to measure price impact, the price change induced by order flow. The Amihud illiquidity ratio for a CDS contract on day t , *Amihud*, is

$$Amihud = \frac{1}{N} \sum_{k=1}^N \frac{|r_k|}{TrdSz_k}, \quad (3)$$

where N is the number of trades on day t , r_k is the percentage price change for the k th trade, i.e., $r_k = 100 \times [(p_k - p_{k-1})/p_{k-1}]$, p_k (p_{k-1}) is the trade price for the k th ($(k-1)$ th) trade, and $TrdSz_k$ is the size of the k th trade (in millions of US dollars). To mitigate the influence of outliers, we only use trades with reported sizes of at least \$1 million. In addition, we compute the natural log of the Amihud ratio, $\log(Amihud)$.

[Jankowitsch, Nashikkar, and Subrahmanyam \(2011\)](#) show that price dispersion (the deviation of traded prices from consensus valuation) is driven by dealer inventory holding costs and investor search costs, two fundamental forces driving liquidity in OTC markets. The price dispersion measure for a CDS contract on day t , *Disp*, is

$$Disp = \sqrt{\frac{1}{\sum_{k=1}^N TrdSz_k} \sum_{k=1}^N TrdSz_k \left(\frac{p_k - EoD_t}{EoD_t} \times 100 \right)^2}, \quad (4)$$

where EoD_t is the CDS contract's consensus valuation on day t , and $TrdSz_k$ and p_k are as previously defined. For a spread-quoted (price-quoted) contract, we use Markit's end-of-day valuation expressed as a spread (price).

5.1.3. Trading activity measures

In addition to trade size (*TrdSz*), we also consider several trading activity measures, which have been shown to be related to transaction costs and liquidity by [Friedwald, Jankowitsch, and Subrahmanyam \(2014\)](#), among others. Daily volume for a CDS contract on day t , *Vol*, is the sum of all trade sizes for that contract on day t . Trade count for a CDS contract on day t , *Ntrds*, is the total number of transactions in that contract on day t . Trading interval, *Intrvl*, refers to the time elapsed between two consecutive trades in a CDS contract measured in days.

5.1.4. Depth-based measures

We measure depth by counting the number of CDS dealers and the number of quotes during the day. For a CDS contract on day t , *Composite depth*, is the number of dealers contributing prices to the calculation of Markit's end-of-day consensus valuation. For a CDS contract on day t , *NQ*, is the total number of quotes disseminated by CDS dealers during the day.

5.1.5. Product characteristics

The literature on fixed income liquidity (e.g., [Friedwald, Jankowitsch, and Subrahmanyam, 2012, 2014](#)) suggests that

maturity and coupon could serve as crude liquidity proxies in the index CDS market. The time-to-maturity of a CDS contract on day t , *Mty*, is calculated as the time (in years) between contract expiration day and trade execution day.²⁵ Index CDS contracts trade with fixed coupons. The coupon of a CDS contract, *Coupon*, is expressed in basis points and is collected from Bloomberg.

5.2. Measures of liquidity in the pre-Dodd-Frank period

To study the liquidity effect of increased post-trade transparency brought about by real-time trade reporting, we also use liquidity measures that do not require high-frequency intraday trade data so that they can be constructed prior to the commencement of real-time trade reporting. While admittedly less sophisticated than the trade-based measures described in the previous section, we believe these non-trade-based liquidity measures can nevertheless capture liquidity changes brought about by increased post-trade transparency.

We draw on CMA intraday quote data, Markit end-of-day composite valuation, and DTCC's Trade Information Warehouse (TIW) weekly reports to construct daily non-trade-based liquidity measures.²⁶ Quoted spread for a CDS contract on day t , *QS*, is now defined as the average across all intraday quoted spreads observed on day t . Relative quoted spread for a CDS contract on day t , *RQS*, is calculated analogously. We also use depth-based measures, *Composite depth* and *NQ*, as liquidity proxies.

In the spirit of the [Amihud \(2002\)](#) measure, we also construct several alternative price impact measures that employ indirect proxies for trading volume, recognizing that the lack of trade size potentially makes these measures noisy proxies for price impact. To the extent that the number of quotes is correlated with trading volume, we construct a price impact measure, *ImpactNQ*, for a CDS contract on day t as $(1/T) \sum_{j=0}^{T-1} (|sc_{t-j}|/NQ_{t-j})$, where NQ_{t-j} is the contract's intraday quotes count on day $t-j$ and $|sc_{t-j}|$ is the absolute percentage change in the CDS contract's Markit composite spread from day $t-j-1$ to day $t-j$, i.e., $|sc_{t-j}| = |100 \times [(csprdt_{t-j} - csprdt_{t-j-1})/csprdt_{t-j-1}]|$ where $csprdt_{t-j-1}$ ($csprdt_{t-j}$) is the composite spread on day $t-j-1$ (day $t-j$). To ensure a reasonable degree of estimation precision, we calculate the price impact measure if at least eight days of data exist in a 60-day estimation window ($T=60$).

Instead of quotes count, the number of dealers can serve as a proxy for trading volume because in the dealer-driven CDS market, liquidity provision by dealers is central to supporting trading activity. This suggests a second price impact measure for a CDS contract on day t , *ImpactDepth*,

²⁵ Note that tenor is the maturity of the index CDS contract when the underlying index series is first created, while time-to-maturity is defined on trade execution day as the time remaining until contract expiration. Thus, over the life of an index CDS contract, tenor remains the same, while time-to-maturity decreases.

²⁶ DTCC operates the TIW, which is a global centralized electronic database of all outstanding CDS contracts. DTCC publishes weekly reports of outstanding positions and aggregate trading activity for single-name and index CDSs on its website (www.dtcc.com).

which is defined in the same way as *ImpactNQ* except that we replace NQ_{t-j} with $Depth_{t-j}$, the CDS contract's *Composite depth* on day $t-j$.

We also construct additional price impact proxies using three measures of aggregate index CDS trading activity from the TIW reports. *ImpactRG* is a price impact measure that uses the gross notional value of index i 's market risk transfer activity (*RG*) as the trading volume proxy, while *ImpactRC* is a price impact measure that uses the number of contracts involved in index i 's market risk transfer activity (*RC*) as the trading volume proxy.²⁷ Following Figlewski (1981) and Bessembinder and Seguin (1993), we use the net notional value of open positions in a CDS index (*ONN*) as a trading activity proxy and construct the price impact measure, *ImpactONN*. Computation of *ImpactRG*, *ImpactRC*, and *ImpactONN* follows that of *ImpactNQ* except that we replace NQ_{t-j} with, respectively, one-fifth of *RG*, *RC*, and *ONN* for the week in which day $t-j$ occurs.²⁸

We construct a modified version of the Jankowitsch, Nashikkar, and Subrahmanyam (2011) price dispersion measure (*Disp*) that only utilizes intraday quotes and end-of-day valuation for the post-trade transparency event study. Thus, a CDS contract's price dispersion on day t is calculated as
$$Disp = \sqrt{1/NQ \sum_{q=1}^{NQ} ((Midquote_q - EoD_t)/EoD_t) \times 100)^2}$$
, where $Midquote_q$ is the midpoint of the q th intraday bid and offer quote on day t , and NQ and EoD_t are as previously defined. To the extent that quote and transaction price dynamics are linked, our measure serves to capture price dispersion in the index CDS market when trading data are unavailable.

5.3. Index CDS trading activity

Our analysis of the final sample begins with an examination of overall trading activity in corporate, municipal, and sovereign index CDSs. Table 3 Panel A reports descriptive statistics of trading activity for the 15 indexes in our sample (see Appendix B for a description of each index). Indexes are ordered based on trade count starting with the index that has the highest number of trades during the sample period. There is considerable variation in trading across indexes. The most active index, CDX NA IG, has 48,254 trades over the course of a year. Furthermore, trading is concentrated in four indexes: CDX NA IG, CDX NA HY, iTraxx Europe, and iTraxx Crossover. These four indexes make up 87% of the trades in our sample. Due to their significant share of overall trading activity, we focus on these key indexes in our post-trade transparency event study and conduct separate analyses for each index when examining the determinants of liquidity. In the index CDS market, the constituents of each index are revised

every six months, resulting in a new series of that index. When the new series is introduced or issued, it becomes the on-the-run (most recent) series and the previous series goes off-the-run along with earlier series. From Panel B of Table 3, we see that 86% of trades are in the on-the-run series. We find that the on-the-run series are more actively traded than the off-the-run series both in terms of trade count and aggregate volume. The higher trading activity in the on-the-run series parallels the findings that recently issued Treasury bonds and corporate bonds attract more trading than similar securities (e.g., Barclay, Hendershott, and Kotz, 2006; Edwards, Harris, and Piwowar, 2007). Panel C of Table 3 presents the distribution of index CDS trades across tenors. The majority of trades are in the 5-year tenor (96% of trades), consistent with the commonly held view that the 5-year tenor attracts the greatest trading interest. The remaining trades are divided among 10-year, 7-year, and 3-year tenors in descending order of importance. The findings on differential trading activity in Table 3 suggest that index identity, on-the-run status, and tenor can delineate segments in the index CDS market with different liquidity characteristics. We address this issue of liquidity segments when reporting summary statistics of liquidity measures in the next section.

5.4. Summary statistics of liquidity measures

In this section, we report summary statistics of liquidity measures for the whole sample. For each combination of index, series, and tenor (e.g., CDX NA IG, Series 19, 5-year tenor), we use all trades executed on a given day to compute daily liquidity measures. We then pool the daily observations across all index-series-tenor combinations and compute summary statistics. The index CDS market uses two quoting conventions: quotes are either in the form of spread or price. Spread quotes are measured in basis points (e.g., a quote of 80 means 80 basis points), while price quotes are measured in percentages (e.g., a quote of 101 means 101% of contract notational amount or par). For each index, dealers post quotes using either one of the conventions. For example, CDX NA IG is a spread-quoted index while CDX NA HY is a price-quoted index. Quoting conventions generate marked differences in price levels and liquidity measures that use transaction price as an input. Hence, for liquidity measures that require transaction prices in their calculations, we compute a separate set of summary statistics for each quoting convention, where $S(P)$ refers to summary statistics for spread-quoted (price-quoted) indexes. Table 4 reports the summary statistics.

The mean *RES* for spread-quoted contracts and price-quoted contracts are 2.73% and 0.27%, respectively. The median *RES* for spread-quoted contracts and price-quoted contracts are 1.37% and 0.18%, respectively. Mean and median trading costs for the two quoting conventions differ by approximately one order of magnitude. These results underscore the importance of performing separate analyses for the two quoting conventions.

Turning to the price impact measure, we find that the mean (median) *Amihud* for spread-quoted contracts is 0.09 (0.02), implying that a \$1 million trade moves the price by 0.09% (0.02%). For price-quoted contracts, both mean and

²⁷ The gross notional value and number of contracts involved in market risk transfer appropriately capture trading activity because they include only trades that change the risk profile of market participants, such as new trades between two parties, the termination of an existing transaction, or the assignment of an existing transaction to a third party.

²⁸ Both *RG* and *ONN* are expressed in billions of US dollars, while *RC* is expressed in thousands of contracts. Because *RG*, *RC*, and *ONN* are weekly variables, we scale them by one-fifth to approximate daily trading activity.

Table 3

Trading activity.

This table reports summary statistics of index CDS trading activity. We summarize 165,919 index CDS trades executed between December 31, 2012 and December 31, 2013 and publicly disseminated by DDR. Panel A reports summary statistics for each index. (See [Appendix B](#) for a description of each index.) Currency is the index's currency denomination. Price type is market quoting convention. S (P) means the index is spread-quoted (price-quoted). We divide indexes into three groups based on the type of bond issuers: corporate, municipal, and sovereign. Within each group, we rank indexes by trade count (i.e., number of trades) starting with the most actively traded index. Pct is an index's trade count as a percentage of all index CDS trades (i.e., as a percentage of 165,919 trades). For each index, we also report the volume transacted (sum of all trade sizes) and summarize the distribution of trade size. Volume and trade size are expressed in millions of contract currency. Panel B reports separate summary statistics for trades in on-the-run and off-the-run index series. For each category, we report the number of trades (Trade count), the number of trades as a percentage of all index CDS trades (Pct), the sum of all trade sizes (Volume), and the average trade size. Volume and average trade size are in millions of US dollars. Panel C reports summary statistics by index tenors. For each tenor, we report the number of trades (Trade count), the number of trades as a percentage of all index CDS trades (Pct), the sum of all trade sizes (Volume), and the average trade size. Volume and average trade size are in millions of US dollars. Tenors are ranked by trade count starting with the most actively traded tenor.

Panel A: Trading activity by index													
	Currency	Price type	Trade count	Pct (%)	Volume (mil)	Trade size (mil)							
						Mean	Median	Mode	Min	Max	S.D.	25th	75th
Corporate													
CDX NA IG	USD	S	48,254	29.08	2,769,226	57	50	100	0.0002	320	40	25	100
CDX NA HY	USD	P	42,568	25.66	869,552	20	13	10	0.0008	320	22	5	25
iTraxx Europe	EUR	S	31,484	18.98	1,185,181	38	25	25	0.0005	240	26	24	50
iTraxx Eur Crossover	EUR	S	22,290	13.43	319,700	14	10	10	0.0010	180	15	5	15
iTraxx Eur Sr Finls	EUR	S	8,464	5.10	212,627	25	23	25	0.0001	150	21	10	28
iTraxx Japan	JPY	S	2,268	1.37	2,561,416	1,129	1,000	1,000	1.0000	11,000	1,465	60	2,000
iTraxx Eur Sub Finls	EUR	S	1,872	1.13	27,254	15	10	25	0.0010	81	17	2	25
iTraxx Asia ex Japan IG	USD	S	1,144	0.69	20,200	18	10	10	0.0050	100	16	10	25
iTraxx Australia	USD	S	736	0.44	16,962	23	15	10	0.0600	110	25	10	25
CDX LatAm Corp	USD	S/P	63	0.04	550	9	5	5	2.0000	39	7	5	10
iTraxx CEEMEA	USD	S	53	0.03	529	10	5	5	1.0000	81	12	5	10
iTraxx Eur HiVol	EUR	S	34	0.02	403	12	10	10	0.2500	57	14	3	10
Municipal													
MCDX	USD	S	760	0.46	19,059	25	25	25	0.1000	110	26	4	28.5
Sovereign													
CDX EM	USD	P	5,721	3.45	76,076	13	10	10	0.0050	110	17	3	18
CDX EM ex-EU	USD	P	208	0.13	2,356	11	5	2	0.0620	100	16	2	12
Panel B: Trading activity in on-the-run and off-the-run index series													
	Trade count			Pct (%)		Volume (US\$mil)			Average trade size (US\$mil)				
On-the-run	143,325			86.38		5,109,128			36				
Off-the-run	22,594			13.62		1,007,431			45				
Panel C: Trading activity across tenors													
Tenor	Trade count			Pct (%)		Volume (US\$mil)			Average trade size (US\$mil)				
5	159,676			96.24		5,672,617			36				
10	4,098			2.47		258,848			63				
7	1,727			1.04		147,997			86				
3	418			0.25		37,098			89				

median estimates are 0.01. Skewness in *Amihud* is considerably reduced by the natural log transformation. Price dispersion is a liquidity measure that reflects inventory holding and search costs in OTC markets. Mean (median) *Disp* for spread-quoted contracts is 2.44 (1.38), implying that the average (median) trade price deviation from end-of-day valuation is 2.44% (1.38%). For price-quoted contracts, the mean and median estimates are 0.20 and 0.13, respectively.

On average, an index contract has a daily volume of \$810 million and trades 22 times a day. Trade size and trade interval for the average index contract are \$43 million and four days, respectively. Trading activity is skewed, however, as the median contract has a daily volume of \$101 million, trades only three times a day, and has a trade

size of \$30 million and a trade interval of half a day. In contrast, measures of market depth tend to be more symmetric. For example, the mean and median *Composite depth* are 11 and ten, respectively. Product characteristics display a fair degree of cross-sectional variation. *Mty* has a mean of 4.6 years and a standard deviation of 1.9 years; *Coupon* has a mean of 238 basis points and a standard deviation of 190 basis points.

5.5. Liquidity effects in different segments and sub-segments of the index CDS market

The analysis of trading activity in [Table 3](#) reveals important differences in trading on the basis of index identity, on-the-run status, and tenor in the index CDS

Table 4

Summary statistics of liquidity measures.

This table provides summary statistics of daily liquidity measures calculated from 165,919 index CDS trades executed between December 31, 2012 and December 31, 2013 and publicly disseminated by DDR. For each combination of index, series, and tenor (e.g., CDX NA IG, Series 19, 5-year tenor) with DDR trade reports, we use all trades executed on a given day to compute daily liquidity measures. We then pool the daily liquidity measures across all index-series-tenor combinations and compute summary statistics. Price type is market quoting convention. S (P) refers to the sample of spread-quoted (price-quoted) indexes. For liquidity measures that are sensitive to quoting convention, we compute summary statistics separately for spread-quoted and price-quoted indexes. *ES* is daily average effective spread. *RES* is daily average relative effective spread. *QS* is daily average quoted spread. *RQS* is daily average relative quoted spread. *Roll* is Roll's (1984) effective spread measure. If quoting convention is spread (price), then quoted and effective bid-ask spreads are in basis points (percentage). Relative quoted and effective bid-ask spreads are in percentage for both price types. *Amihud* is the [Amihud \(2002\)](#) price impact measure (in percentage). *Log(Amihud)* is the natural log of Amihud. *Disp* is the [Jankowitsch, Nashikkar, and Subrahmanyam \(2011\)](#) price dispersion measure (in percentage). *Vol* is daily trading volume (in US\$ millions). *Ntrds* is daily trade count. *TrdSz* is daily average trade size (in US\$ millions). *Intrvl* is daily average time elapsed between consecutive trades (in days). *Mty* is time between index expiration day and trade execution day (in years). *Coupon* is index series coupon (in basis points). *Composite depth* is the number of CDS dealers contributing prices for end-of-day consensus valuation. *NQ* is quotes count.

	Price type	NOBS	Mean	Median	SD	Min	Max	Skewness	Kurtosis
Transaction cost									
<i>ES</i>	S	3,979	3.37	1.71	7.74	0.00	153.00	10.52	152.49
	P	1,436	0.29	0.19	0.40	0.00	5.29	5.71	45.12
<i>RES</i>	S	3,979	2.73	1.37	4.62	0.00	52.91	5.14	34.33
	P	1,436	0.27	0.18	0.38	0.00	4.86	5.64	43.95
<i>QS</i>	S	3,979	2.37	2.00	1.91	0.42	20.00	2.42	11.02
	P	1,436	0.23	0.22	0.10	0.00	0.80	0.78	0.31
<i>RQS</i>	S	3,979	2.23	1.51	2.46	0.38	24.58	3.49	16.26
	P	1,436	0.21	0.21	0.10	0.00	0.77	0.77	0.39
<i>Roll</i>	S	2,531	2.60	1.00	5.55	0.00	93.08	7.57	91.24
	P	854	0.18	0.11	0.22	0.00	2.23	2.78	13.08
Price impact and dispersion									
<i>Amihud</i>	S	3,889	0.09	0.02	0.53	0.00	17.28	26.51	817.78
	P	1,217	0.01	0.01	0.04	0.00	0.95	16.26	351.49
<i>Log(Amihud)</i>	S	3,099	−3.30	−3.32	1.37	−9.47	2.85	0.09	0.84
	P	970	−4.85	−4.81	1.30	−15.19	−0.06	−1.28	8.34
<i>Disp</i>	S	5,838	2.44	1.38	3.30	0.00	24.96	3.43	14.44
	P	1,709	0.20	0.13	0.23	0.00	2.30	3.68	19.75
Trading activity									
<i>Vol</i>		7,547	810.46	101.39	2153.68	0.00	32086.65	4.99	34.03
<i>Ntrds</i>		7,547	21.98	3.00	49.59	1.00	552.00	3.71	17.92
<i>TrdSz</i>		7,547	42.52	29.68	40.14	0.00	327.41	2.16	8.33
<i>Intrvl</i>		7,428	4.29	0.52	15.07	0.00	274.03	8.19	89.44
Product characteristics									
<i>Mty</i>		7,547	4.56	4.71	1.90	0.02	10.26	0.98	2.31
<i>Coupon</i>		7,547	238.24	100.00	189.65	40.00	975.00	0.67	−1.43
Depth									
<i>Composite depth</i>		7,547	10.60	10.00	3.20	3.00	17.00	−0.11	−1.08
<i>NQ</i>		5,633	81.18	60.00	76.59	1.00	356.00	0.73	−0.55

market. Given the intimate relation between trading and liquidity, we investigate whether index identity, on-the-run status, and tenor identify segments exhibiting different liquidity characteristics. We begin by exploring liquidity variations across indexes. As before, we calculate daily liquidity measures for each index-series-tenor combination. Then, we pool these measures across all series-tenor combinations within each index (e.g., all combinations of series and tenor of CDX NA IG) and compute the average daily liquidity measures for the index. Because CDX LatAm Corp changed its quoting convention from price to spread when moving from Series 18 to 19, we treat these series as separate indexes in our calculation and reporting. We organize indexes by quoting convention and report their average daily liquidity measures in [Table 5](#), Panel A. There is substantial cross-sectional variation in liquidity among the indexes, that is, index identity delineates market segments with different levels of liquidity. For instance, in the spread-quoted sample, *Amihud* ranges from 0.06 (CDX NA

IG) to 0.36 (CDX CEEMEA). Price-quoted indexes also exhibit noticeable differences in liquidity. For example, daily quotes count can be as low as five for CDX EM ex-EU and as high as 72 for CDX NA HY.

Next, we consider on-the-run status as a source of liquidity variation by comparing the liquidity of the on-the-run and off-the-run sub-segments. The analysis focuses on CDX NA IG (IG), CDX NA HY (HY), iTraxx Europe (Eur), and iTraxx Europe Crossover (Xover) because [Table 3](#), Panel A shows that these indexes dominate trading in the index CDS market. For each index, we assign daily liquidity measures at the index-series-tenor level to either the on-the-run or off-the-run sub-segment based on whether the series is the most recent or not. We then compute the average liquidity measures for each sub-segment and conduct *t*-tests of differences in means. We find strong evidence that the on-the-run sub-segment exhibits higher liquidity ([Table 5](#), Panel B). Across all four indexes, on-the-run trades have lower transaction costs as measured by *ES*, *RES*,

Table 5

Liquidity variations across segments.

This table reports mean daily liquidity by segments in the index CDS market. The data consist of 165,919 index CDS trades executed between December 31, 2012 and December 31, 2013 and publicly disseminated by DDR. For each combination of index, series, and tenor (e.g., CDX NA IG, Series 19, 5-year tenor) with DDR trade reports, we use all trades executed on a given day to compute daily liquidity measures. See Table 4 for definitions of daily liquidity measures. In Panel A, we pool the daily liquidity measures across all series-tenor combinations within each index and compute the mean daily liquidity measures for the index. Because CDX LatAm Corp changed its quoting convention from price to spread when moving from Series 18 to 19, we treat these series as separate indexes in our calculation and reporting. Columns 1–4 (5–16) report mean daily liquidity measures for price-quoted (spread-quoted) indexes. In Panel B, we examine daily liquidity measures in the on-the-run and off-the-run segments of CDX NA IG (IG), CDX NA HY (HY), iTraxx Europe (Eur), and iTraxx Europe Crossover (Xover). For each index, we assign daily liquidity measures to the on-the-run (off-the-run) segment if the index series is the most recent (is not the most recent) on a given day. Column On (Off) reports mean liquidity for the on-the-run (off-the-run) segment. The column On-off reports the difference in mean liquidity between the two segments. In Panel C, we examine daily liquidity measures in the 5-year and non-5-year segments of CDX NA IG (IG) and iTraxx Europe (Eur). For each index, we assign daily liquidity measures to the 5-year (non-5-year) segment based on whether the index tenor is 5 years (is not 5 years). Column 5Y (Non5Y) reports mean liquidity for the 5-year (non-5-year) segment. The column 5Y-non5Y reports the difference in mean liquidity between the two segments. In Panels B and C, we test the difference in mean liquidity between segments using a two-sample *t*-test. *, **, *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Liquidity across indexes

	Price type: Price				Price type: Spread											
	(1) NA HY	(2) LatAm Corp 18	(3) EM	(4) EM ex-EU	(5) NA IG	(6) Eur	(7) Eur Xover	(8) Eur Sr Finls	(9) Eur Sub Finls	(10) Eur HiVol	(11) Japan	(12) Asia ex-JP IG	(13) Australia	(14) LatAm Corp 19	(15) CEEMEA	(16) MCDX
Transaction cost																
ES	0.30	0.30	0.27	0.09	1.99	2.44	8.30	3.17	5.47	4.35	2.82	2.68	1.56	6.00	6.08	4.06
RES	0.28	0.27	0.25	0.08	3.08	3.14	2.41	2.33	2.53	2.94	2.89	2.12	1.34	2.45	2.38	2.92
QS	0.25	0.39	0.16	0.17	1.60	1.37	3.15	1.57	5.68	5.92	1.97	2.30	1.50	10.00	6.44	5.21
RQS	0.24	0.36	0.14	0.15	2.90	2.07	1.00	1.19	2.71	3.86	2.05	1.80	1.29	4.08	2.62	3.67
Roll	0.19	0.54	0.16	0.10	1.22	2.00	6.79	2.64	3.10	–	2.22	2.59	1.62	–	5.72	2.62
Price impact and dispersion																
Amihud	0.01	0.02	0.01	0.01	0.06	0.08	0.10	0.07	0.12	0.03	0.16	0.16	0.08	0.12	0.36	0.20
Log(Amihud)	–4.84	–4.09	–4.90	–4.86	–3.75	–3.78	–2.78	–3.12	–2.54	–2.03	–2.50	–2.65	–3.22	–1.48	–1.75	–2.89
Disp	0.21	0.24	0.18	0.11	2.62	2.65	2.39	3.12	2.29	1.84	2.05	1.54	1.06	3.95	2.00	1.55
Trading activity																
Vol	764.78	13.86	158.49	36.82	1529.96	1040.94	779.95	573.32	98.57	18.46	84.57	82.11	84.81	12.46	23.00	64.83
Ntrds	37.44	1.61	11.92	3.25	26.66	20.84	40.97	17.20	5.11	1.17	7.32	4.65	3.68	1.38	2.30	2.59
TrdSz	30.47	9.60	12.70	12.14	65.58	65.85	23.81	34.75	23.93	17.32	14.90	17.24	20.37	7.97	11.97	28.29
Intrvl	3.33	3.02	1.95	3.45	3.85	6.16	2.92	5.79	5.48	14.54	3.70	0.86	0.97	6.97	8.41	8.64
Product characteristics																
Mty	3.57	4.81	4.56	4.75	4.60	4.65	4.53	4.52	4.55	4.83	4.82	4.96	5.00	5.10	4.89	6.77
Coupon	495.23	500.00	500.00	500.00	93.38	116.38	503.96	101.35	446.72	100.00	100.00	100.00	100.00	500.00	100.00	99.56
Depth																
Composite depth	10.39	5.96	10.37	7.55	10.44	11.32	13.43	10.92	10.40	9.07	12.25	9.37	9.73	5.54	5.17	5.39
NQ	71.88	13.33	55.73	4.78	60.72	104.06	157.59	138.93	136.33	96.85	31.65	52.03	60.63	3.50	15.95	30.67

Panel B: On-the-run vs. off-the-run

	IG			HY			Eur			Xover		
	On	Off	On-off	On	Off	On-off	On	Off	On-off	On	Off	On-off
Transaction cost												
ES	1.267	2.324	–1.057***	0.197	0.334	–0.137***	1.477	3.277	–1.800***	4.621	13.230	–8.609***
RES	1.409	3.852	–2.443***	0.188	0.314	–0.126***	1.418	4.620	–3.202***	1.145	4.110	–2.965***
QS	1.077	1.839	–0.762***	0.142	0.286	–0.144***	0.766	1.891	–1.125***	2.300	4.290	–1.989***
RQS	1.195	3.689	–2.494***	0.135	0.268	–0.133***	0.749	3.201	–2.452***	0.577	1.577	–1.000***
Roll	1.151	1.255	–0.104	0.216	0.172	0.043**	1.518	2.500	–0.982***	5.894	8.996	–3.102*

(continued on the next page)

Table 5 (Continued)

Panel B: on-the-run vs. off-the-run												
	IG			HY			Eur			Xover		
	On	Off	On-off	On	Off	On-off	On	Off	On-off	On	Off	On-off
Price impact and dispersion												
Amihud	0.042	0.066	−0.024**	0.012	0.015	−0.003	0.048	0.099	−0.051	0.076	0.141	−0.065***
Log(Amihud)	−3.818	−3.715	−0.103	−4.672	−4.945	0.273***	−3.825	−3.750	−0.075	−2.817	−2.718	−0.099
Disp	1.478	3.021	−1.544***	0.259	0.196	0.062***	1.870	3.027	−1.157***	1.541	3.123	−1.582***
Trading activity												
Vol	4,829.96	375.821	4454.139***	2,853.709	172.992	2680.717***	2,673.476	235.962	2437.514***	1,497.042	156.493	1340.550***
Ntrds	86.192	5.839	80.353***	146.112	6.652	139.459***	55.281	3.853	51.428***	80.814	6.337	74.477***
TrdSz	58.467	68.066	−9.599***	19.150	33.671	−14.521***	59.036	69.205	−10.169***	18.463	28.451	−9.989***
Intrvl	1.382	4.723	−3.341***	0.015	4.278	−4.263***	1.527	8.481	−6.954***	0.032	5.480	−5.448***
Product characteristics												
Mty	6.466	3.954	2.512***	4.984	3.172	1.812***	6.001	3.989	2.012***	5.003	4.118	0.885***
Coupon	100.000	91.059	8.941***	500.000	493.877	6.123***	100.000	124.452	−24.452***	500.000	507.405	−7.405**
Depth												
Composite depth	11.949	9.914	2.035***	12.163	9.885	2.278***	12.170	10.902	1.268***	15.324	11.790	3.534***
NQ	115.120	36.234	78.886***	160.892	44.863	116.029***	157.997	59.250	98.747***	191.593	114.357	77.236***
Panel C: 5-Year vs. non-5-year												
	IG			Eur								
	5Y	Non5Y	5Y-non5Y	5Y	Non5Y	5Y-non5Y						
Transaction cost												
ES	1.516	2.580	−1.064***	2.030	3.450	−1.420***						
RES	2.548	3.744	−1.196***	2.384	4.973	−2.589***						
QS	1.160	2.143	−0.983***	0.832	2.679	−1.847***						
RQS	2.367	3.568	−1.201***	1.103	4.408	−3.305***						
Roll	1.203	1.237	−0.034	1.774	2.438	−0.665						
Price impact and dispersion												
Amihud	0.071	0.047	0.024*	0.041	0.118	−0.076						
Log(Amihud)	−3.506	−4.096	0.590***	−3.727	−3.866	0.14						
Disp	2.378	2.842	−0.464***	2.326	2.859	−0.532***						
Trading activity												
Vol	2,918.600	275.656	2,642.944***	2,323.921	181.836	2,142.085***						
Ntrds	52.133	3.651	48.482***	48.183	2.525	45.658***						
TrdSz	55.029	75.108	−20.079***	52.935	74.493	−21.558***						
Intrvl	2.250	5.326	−3.076***	3.083	8.272	−5.189***						
Product characteristics												
Mty	4.018	5.135	−1.118***	4.299	4.891	−0.593***						
Coupon	100.984	86.504	14.480***	101.691	126.210	−24.519***						
Depth												
Composite depth	11.540	9.449	2.091***	13.403	9.927	3.476***						
NQ	103.352	8.378	94.974***	149.415	3.019	146.396***						

QS, and RQS. On-the-run trades also have greater market depth as measured by *Composite depth* and *NQ* and higher trading activity as measured by *Vol*, *Ntrds*, and *Intrvl*. In addition, *Roll*, *Amihud*, and *Disp* tend to be lower for on-the-run trades.

For the last part of our sub-segment analysis, we turn our attention to tenor. Specifically, we explore liquidity differences between the 5-year and non-5-year sub-segments of IG and Eur. We focus on these two indexes because a sizable fraction of each index's daily sample consists of trades in non-5-year tenors.²⁹ For each index, we assign daily liquidity measures at the index-series-tenor level to the 5-year sub-segment if the tenor is five years; liquidity measures for 3-, 7-, or 10-year tenors are assigned to the non-5-year sub-segment. We compute the average liquidity measures for each sub-segment and conduct *t*-tests of differences in means. The results in Panel C of Table 5 indicate that liquidity is higher in the 5-year sub-segment. For both IG and Eur, we find that 5-year trades have lower transaction costs as measured by *ES*, *RES*, *QS*, and *RQS*, as well as lower price dispersion. In addition, the 5-year sub-segment has greater market depth and more active trading as evidenced by the larger *Composite depth*, *NQ*, *Vol*, *Ntrds*, and smaller *Intrvl*.

6. Empirical results

6.1. Does the Dodd-Frank Act affect liquidity?

Hypothesis 1. Index CDS liquidity improves after the start of real-time reporting and public dissemination of index CDS trades.

The first hypothesis we test is whether liquidity improves after the start of real-time reporting and public dissemination of index CDS trades on December 31, 2012. Using December 31, 2012 as the event day (day 0), we compare index CDS liquidity in the pre-event and event periods. As discussed in Section 5.2, we use liquidity measures that do not require high-frequency intraday trade data, which are unavailable prior to the commencement of the Dodd-Frank trade reporting regime. The pre-event (event) period liquidity measure is the daily average liquidity measure calculated from day –440 to day –21 (day 5 to day 20).³⁰ To ensure that our results are not driven by cross-sectional liquidity differences across indexes, on-the-run status, and index tenor, we conduct a separate event

study of the 5-year, on-the-run series of IG, HY, Eur, and Xover. For each of these indexes, we report liquidity measures in the pre-event and event periods, their difference as well as the result of the two-sample *t*-test in Table 6.

We find evidence that index CDS liquidity improves after the commencement of the reporting and public dissemination of index CDS trades. *QS*, *RQS*, and *Disp*, which are inverse measures of liquidity, register statistically significant declines in the event period across all four indexes. For example, the average *RQS* of IG falls from 0.64% in the pre-event period to 0.56% in the event period, and the decline is statistically significant at the 1% level. *Disp*, which captures dealer inventory and search costs, exhibits statistically significant decreases with magnitudes ranging from –0.19% to –0.55%. The various price impact measures demonstrate an overall pattern of decline in the event period, which is consistent with liquidity improvement after the start of the Dodd-Frank trade reporting regime. For instance, price impact as measured by *ImpactONN* exhibits statistically significant reductions in the event period for all indexes. We also find evidence of liquidity improvement using *Composite depth*, a depth-based measure of liquidity. The increases in *Composite depth* range from 0.45 to 1.75, all of which are statistically significant.

The efficacy of the Dodd-Frank Act's OTC derivatives reforms has been widely debated although there has been little empirical evidence to date. Our event study of the liquidity effects of Dodd-Frank mandated post-trade reporting represents a first step in addressing this gap in the literature. Our evidence suggests that the increased post-trade transparency associated with the public dissemination of OTC derivatives trades improves liquidity in the index CDS market. Apart from trade reporting and dissemination, the Dodd-Frank Act also mandates central clearing, establishes SEFs, and provides regulatory recognition of derivatives end-users and customized derivatives trades. In the remainder of this paper, we examine the liquidity effects of these post-crisis reforms using our high-frequency data set of index CDS trades and quotes.

6.2. The Dodd-Frank reforms and index CDS liquidity

Hypotheses 2.1–2.5. Transaction costs are lower and liquidity is higher for centrally cleared trades, trades executed on SEFs, trades with end-users, bespoke trades, and block trades.

The second hypothesis that we explore is how transaction costs and liquidity are affected by the deal characteristics that came to be defined in the post-Dodd-Frank era. We first investigate their effects on daily liquidity measures, and then conduct transaction-level analyses to take advantage of the high granularity of the real-time trade reports.

6.2.1. Regression of daily liquidity measures on Dodd-Frank deal characteristics

We estimate daily regressions of liquidity measures on the Dodd-Frank deal characteristics and control variables. Following Friewald, Jankowitsch, and Subrahmanyam (2014), we focus on the relative effective spread, the

²⁹ For IG, 47% (53%) of its daily sample consists of 5-year (non-5-year) trades. For Eur, 40% (60%) of its daily sample consists of 5-year (non-5-year) trades. In contrast, HY and Xover have predominantly 5-year trades in their daily samples, making these indexes less suitable candidates for the analysis. Trades in 5-year contracts make up 92% of HY's daily sample, while that fraction is close to 100% in Xover's daily sample.

³⁰ DTCC's TIW first reported outstanding positions data (including outstanding net notional, *ONN*) of untranch index CDSs for the week that ended on Friday, April 29, 2011. We require *ONN* to construct *ImpactONN*, a daily price impact measure that uses one-fifth of the weekly *ONN*. Hence, we set the pre-event period to begin on day –440, which is the Monday that falls on the same week as Friday, April 29, 2011. The event period starts on day 5 to allow market participants time to get used to the new reporting regime.

Table 6

Does Dodd-Frank affect liquidity?

We examine index CDS liquidity around the commencement of post-trade reporting on December 31, 2012 by comparing liquidity in the pre-event (day –440 to day –21) and event (day 5 to day 20) periods. The sample consists of 5-year, on-the-run series of CDX NA IG (IG), CDX NA HY (HY), iTraxx Europe (Eur), and iTraxx Europe Crossover (Xover). For each index, Pre-event (Event) is mean daily liquidity in the pre-event (event) period. Δ is Event minus Pre-event. *QS* is quoted spread. *RQS* is relative quoted spread. *ImpactNQ*, *ImpactDepth*, *ImpactRG*, *ImpactRC*, and *ImpactONN* are price impact measures where trading activity is proxied by, respectively, quotes count, composite depth, gross notional value of market risk transfer, number of contracts involved in market risk transfer, and net notional value of open positions. $\text{Log}(\cdot)$ is the natural log operator. Notional values are in billions of US dollars. Contracts are in thousands. *Disp* is price dispersion. *Composite depth* is the number of CDS dealers contributing prices for end-of-day consensus valuation. *NQ* is quotes count. We test the difference in mean liquidity using a two-sample *t*-test. *, **, *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	IG			HY			Eur			Xover		
	Pre-event	Event	Δ	Pre-event	Event	Δ	Pre-event	Event	Δ	Pre-event	Event	Δ
Bid-ask spread												
<i>QS</i>	0.701	0.489	–0.213***	0.204	0.125	–0.079***	0.880	0.554	–0.326***	3.443	2.215	–1.229***
<i>RQS</i>	0.638	0.562	–0.076***	0.213	0.122	–0.091***	0.592	0.534	–0.058***	0.564	0.522	–0.042***
Price impact and dispersion												
<i>ImpactNQ</i>	0.011	0.010	–0.0004*	0.012	0.011	–0.001***	0.011	0.008	–0.003***	0.010	0.008	–0.002***
<i>ImpactDepth</i>	0.145	0.128	–0.017***	0.183	0.149	–0.034***	0.180	0.113	–0.068***	0.166	0.111	–0.054***
<i>ImpactRG</i>	0.105	0.077	–0.028*	0.525	0.408	–0.116***	0.118	0.121	0.003	0.421	0.463	0.042***
<i>ImpactRC</i>	6.857	4.917	–1.941**	8.134	5.512	–2.622***	6.398	5.777	–0.620***	6.570	6.234	–0.336***
<i>ImpactONN</i>	0.137	0.121	–0.015***	0.524	0.406	–0.118***	0.275	0.192	–0.084***	0.938	0.677	–0.261***
$\text{Log}(\text{ImpactNQ})$	–4.581	–4.570	0.011	–4.494	–4.531	–0.037	–4.536	–4.775	–0.239***	–4.595	–4.779	–0.184***
$\text{Log}(\text{ImpactDepth})$	–1.972	–2.060	–0.087***	–1.754	–1.908	–0.154***	–1.735	–2.185	–0.449***	–1.819	–2.197	–0.378***
$\text{Log}(\text{ImpactRG})$	–2.430	–2.562	–0.132***	–0.694	–0.899	–0.205***	–2.184	–2.115	0.069***	–0.908	–0.771	0.137***
$\text{Log}(\text{ImpactRC})$	1.783	1.591	–0.193***	2.064	1.705	–0.359***	1.822	1.754	–0.068***	1.848	1.829	–0.018
$\text{Log}(\text{ImpactONN})$	–2.026	–2.111	–0.085***	–0.721	–0.904	–0.183***	–1.334	–1.655	–0.321***	–0.118	–0.392	–0.274***
<i>Disp</i>	1.194	0.860	–0.334***	0.326	0.138	–0.188***	1.425	0.873	–0.552***	1.310	0.760	–0.550***
Depth												
<i>Composite depth</i>	12.883	13.333	0.450**	10.973	11.733	0.761***	12.835	14.125	1.290***	13.123	14.875	1.752***
<i>NQ</i>	180.091	192.333	12.243	173.081	176.267	3.186	210.988	211.000	0.012	211.819	211.625	–0.194

Amihud (2002) price impact measure, the Roll (1984) implied effective spread, and the Jankowitsch, Nashikkar, and Subrahmanyam (2011) price dispersion measure. The daily liquidity measures are constructed as follows. For each combination of index, series, and tenor (e.g., CDX NA IG, Series 19, 5-year tenor) with DDR trade reports, we sort all trades executed on a given day into buckets based on the five Dodd-Frank determinants extracted from the DDR trade reports: (i) whether the trade is centrally cleared, (ii) whether the trade is executed on an SEF or designated contract market, (iii) whether the trade is an end-user trade, (iv) whether the trade is bespoke, and (v) whether the trade is a block trade. For each bucket (e.g., all centrally cleared, non-end-user, non-bespoke, non-block trades executed away from an SEF or designated contract market), we use all assigned trades on a given day to compute daily *RES*, $\text{Log}(\text{Amihud})$, *Disp*, and *Roll*. Linked to each daily liquidity measure are five dummy variables each indicating whether a particular Dodd-Frank determinant characterizes the trades used to compute the liquidity measure. *Clear* equals one if the trade is centrally cleared, and equals zero otherwise. *Venue* equals one if the trade is executed on a swap execution facility or designated contract market, and equals zero otherwise. *Eu* equals one for an end-user trade, and equals zero otherwise. *Bespoke* equals one for a bespoke trade, and equals zero otherwise. *Block* equals one if the trade is reported as a block trade, and equals zero otherwise.

The control variables capture liquidity effects associated with segments, trading activity, and product characteristics. The results in Table 5 demonstrate that index identity, on-the-run status, and tenor delineate index CDS mar-

ket segments exhibiting different liquidity characteristics. To account for liquidity variations across different indexes, we include index dummies. To control for liquidity differences between on-the-run and off-the-run trades, we include the dummy variable, *OTR*, which equals one if the index series is the most recent series, and equals zero otherwise. Because 5-year index trades tend to be more liquid, we include *Tenor5*, which equals one if the index tenor is five years, and equals zero otherwise. Following Friewald, Jankowitsch, and Subrahmanyam (2014), we include trading activity variables and product characteristics that are related to liquidity in the OTC fixed income market. We expect $\text{Log}(\text{Vol})$, *Ntrds*, *Composite depth*, and *NQ* to be positively related to liquidity because liquid securities tend to be more actively traded. In contrast, we expect *Intrvl* to be inversely related to liquidity because illiquid securities are more likely to experience long periods without trading activity. We expect $\text{Log}(\text{TrdSz})$ to be inversely related to liquidity, as suggested by studies of liquidity in other fixed income markets. To account for liquidity variations due to product characteristics, we include maturity (*Mty*) and coupon (*Coupon*). For each liquidity measure, we estimate separate regressions for spread-quoted and price-quoted contracts. We calculate statistical significance using heteroskedasticity-consistent standard errors clustered at the index level. Table 7 reports the daily regression results.

The regression results strongly support the hypotheses that transaction costs are lower and liquidity is higher for centrally cleared trades, trades executed on SEFs, end-user trades, and bespoke trades. The coefficient on *Clear* is negative and statistically significant in virtually all instances.

Table 7

Dodd-Frank reforms and index CDS liquidity.

This table reports regressions of daily index CDS liquidity on the Dodd-Frank determinants while controlling for the effects of segments, trading activity, and product characteristics. The data consist of 165,919 index CDS trades executed between December 31, 2012 and December 31, 2013 and publicly disseminated by DDR. For each combination of index, series, and tenor (e.g., CDX NA IG, Series 19, 5-year tenor) with DDR trade reports, we sort all trades executed on a given day into buckets based on five Dodd-Frank determinants extracted from DDR trade reports: (i) whether the trade is centrally cleared, (ii) whether the trade is executed on a swap execution facility or designated contract market, (iii) whether the trade is an end-user trade, (iv) whether the trade is bespoke, and (v) whether the trade is a block trade. For each bucket (e.g., all centrally cleared, non-end-user, non-bespoke, non-block trades executed away from a swap execution facility or designated contract market), we use all assigned trades on a given day to compute daily liquidity measures. The daily liquidity measures are the relative effective spread (*RES*), the natural log of the Amihud (2002) measure ($\text{Log}(\text{Amihud})$), the Jankowitsch, Nashikkar, and Subrahmanyam (2011) price dispersion measure (*Disp*), and Roll's (1984) effective spread measure (*Roll*). Explanatory variables are constructed from DDR trade reports and capture various Dodd-Frank reforms and the effects of segments, trading activity, and product characteristics. *Clear* equals one if the trade is centrally cleared and zero otherwise. *Venue* equals one if the trade is executed on a swap execution facility or designated contract market and zero otherwise. *Eu* equals one for an end-user trade and zero otherwise. *Bespoke* equals one for a bespoke trade and zero otherwise. *Block* equals one if the trade is reported as a block trade and zero otherwise. *OTR* equals one if the index series is the most recent series, and equals zero otherwise. *Tenor5* equals one if the index tenor is 5 years, and equals zero otherwise. $\text{Log}(\text{Vol})$ is natural log of daily volume. *Ntrds* is daily trade count. $\text{Log}(\text{TrdSz})$ is natural log of daily average trade size. Both daily volume and daily average trade size are in millions of US dollars. *Composite depth* is the number of dealers contributing prices to the calculation of Markit's end-of-day consensus valuation. *NQ* is daily quotes count. *Intrvl* is the daily average time elapsed between consecutive trades (in days). *Mty* is time between index expiration day and trade execution day (in years). *Coupon* is index series coupon (in basis points). The columns labeled S (P) report regression results based on spread-quoted (price-quoted) contracts. All regressions include index dummies whose coefficients are suppressed for brevity. Numbers in parentheses are heteroskedasticity-consistent standard errors clustered at the index level. *N* is the number of observations. *, **, *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	<i>RES</i>		$\text{Log}(\text{Amihud})$		<i>Disp</i>		<i>Roll</i>	
	S	P	S	P	S	P	S	P
<i>Clear</i>	−0.445*** (0.092)	−0.079*** (0.009)	−0.211*** (0.050)	−0.144* (0.051)	−0.289*** (0.042)	−0.056*** (0.006)	−0.286*** (0.076)	−0.021 (0.011)
<i>Venue</i>	−0.301*** (0.081)	−0.090*** (0.011)	−0.325*** (0.057)	−0.409*** (0.022)	−0.023 (0.100)	−0.037** (0.008)	−0.469 (0.311)	−0.042*** (0.001)
<i>Eu</i>	−0.625*** (0.072)	−0.063*** (0.008)	−0.144*** (0.043)	−0.065 (0.031)	−0.229*** (0.035)	−0.036*** (0.002)	−0.226* (0.110)	−0.037*** (0.002)
<i>Bespoke</i>	−0.509*** (0.094)	−0.083*** (0.004)	−0.392*** (0.091)	−0.646* (0.212)	−0.180** (0.080)	−0.041*** (0.001)	−0.397** (0.172)	−0.061*** (0.006)
<i>Block</i>	0.276*** (0.087)	−0.073*** (0.009)	0.016 (0.052)	0.015 (0.018)	0.234*** (0.053)	−0.045*** (0.007)	0.636** (0.232)	−0.018** (0.004)
<i>OTR</i>	−1.182*** (0.171)	−0.093** (0.027)	−0.414*** (0.075)	−0.293* (0.093)	−0.781*** (0.105)	−0.044* (0.018)	−1.327*** (0.404)	−0.058*** (0.005)
<i>Tenor5</i>	−1.503*** (0.077)	−0.063*** (0.003)	−0.484*** (0.053)	0.559** (0.098)	−0.944*** (0.046)	0.014 (0.010)	−0.188 (0.385)	0.117*** (0.010)
$\text{Log}(\text{Vol})$	−0.056 (0.046)	−0.015 (0.012)	0.085** (0.033)	0.081 (0.048)	0.132*** (0.029)	−0.001 (0.014)	0.372** (0.155)	0.004 (0.009)
<i>Ntrds</i>	0.008*** (0.002)	0.001*** (0.000)	0.007*** (0.001)	0.004*** (0.001)	0.004** (0.001)	0.001** (0.000)	−0.000 (0.002)	0.001** (0.000)
$\text{Log}(\text{TrdSz})$	−0.126 (0.113)	0.019 (0.018)	−1.075*** (0.050)	−0.959*** (0.088)	−0.185*** (0.050)	0.009 (0.017)	−0.486 (0.267)	0.003 (0.016)
<i>Composite depth</i>	0.007 (0.066)	−0.010* (0.003)	−0.027*** (0.005)	−0.023** (0.005)	0.038** (0.017)	−0.008** (0.002)	−0.033 (0.035)	−0.005 (0.004)
<i>NQ</i>	−0.003*** (0.001)	0.000 (0.000)	0.001** (0.000)	0.004*** (0.000)	0.000 (0.000)	0.001** (0.000)	0.003*** (0.001)	0.001** (0.000)
<i>Intrvl</i>	0.028*** (0.005)	−0.003** (0.001)	0.004 (0.021)	−0.060*** (0.002)	0.006 (0.004)	−0.001** (0.000)	0.106 (0.527)	−0.053 (0.023)
<i>Mty</i>	−0.534*** (0.042)	0.003 (0.009)	−0.174*** (0.017)	−0.203*** (0.024)	−0.323*** (0.011)	0.012 (0.014)	0.025 (0.041)	−0.018 (0.026)
<i>Coupon</i>	−0.002 (0.006)	−0.001*** (0.000)	0.001 (0.002)	−0.003*** (0.000)	−0.008*** (0.003)	−0.000*** (0.000)	0.008*** (0.002)	−0.000 (0.000)
Constant	7.757*** (1.081)	1.023*** (0.008)	1.817*** (0.224)	−0.771*** (0.014)	5.768*** (0.650)	0.457*** (0.022)	3.019*** (0.246)	0.325** (0.042)
Index dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	12,017	4,598	9,406	3,573	12,407	4,671	7,122	2,737
Adjusted <i>R</i> ²	0.1092	0.0506	0.4460	0.3249	0.0902	0.0681	0.1073	0.0550

The relative effective spread in the spread-quoted (price-quoted) sample is lower by 0.45% (0.08%) for cleared trades compared to noncleared trades. The Roll estimate of effective spread is also lower for cleared trades, although the difference is statistically significant only in the spread-quoted sample. Both price impact and price dispersion are lower for cleared trades compared to noncleared trades and the differences are always statistically significant. For

example, price dispersion is 0.29% (0.06%) lower for cleared trades in the spread-quoted (price-quoted) sample.

Trades executed on SEFs have lower relative effective spread and lower price impact in both spread-quoted and price-quoted samples. In addition, price-quoted trades executed on SEFs also have smaller price dispersion and Roll implied effective spread. The results for *Eu* and *Bespoke* show that end-user and bespoke trades have

consistently lower trading cost, price impact, and price dispersion. Findings for block trades are inconclusive. Trading cost and price dispersion are higher for blocks in the spread-quoted sample, but lower for blocks in the price-quoted sample. In terms of price impact, there is no discernible difference between blocks and non-blocks.

In line with the segment analysis of Table 5, we find that on-the-run trades and 5-year trades are more liquid. For example, the coefficient on *OTR* is negative and statistically significant in all regression specifications. We also find some evidence that index level effects help explain liquidity variations. Specifically, CDX NA IG, iTraxx Europe, iTraxx Europe Senior Financials, and iTraxx Australia tend to exhibit higher liquidity than other indexes in the spread-quoted sample. For the sake of brevity, coefficients on index dummies are not reported in this paper. Trading activity variables and product characteristics are statistically significant in many cases, underscoring the need to control for liquidity effects associated with these variables.

6.2.2. Regression analysis of transaction-level trading costs

In this section, we exploit the high granularity of our data set to examine how the Dodd-Frank determinants relate to trading costs at the transaction level. We measure CDS trading cost as the relative effective spread because transactions can occur within or outside quoted bid and offer prices due to negotiations between customers and dealers. These negotiations are driven by price-affecting factors, which in turn are influenced by the Dodd-Frank reforms. In the transaction-level regressions of relative effect spread, the explanatory variables of interest include *Clear*, *Venue*, *Eu*, *Bespoke*, and *Block*, all of which have been defined in the previous section. When the indication of end-user status is missing for a trade report, we set *Eu* to zero, and we set a missing end-user status dummy variable, *Misseu*, to one. Trades with end-user status data have *Misseu* set to zero.³¹ More importantly, we introduce event-based dummy variables that provide additional tests to more precisely establish the links between various Dodd-Frank deal characteristics and trading costs. Use of SEFs surged after their official start date of October 2, 2013 (Fig. 2, Panel C), suggesting that liquidity variations related to trading venue could be more pronounced after that date. Accordingly, we add a dummy variable *Sefdt*, which equals one if the trade date occurs on or after October 2, 2013, and equals zero otherwise. More importantly, we interact *Venue* and *Sefdt* to capture liquidity effects occurring after the SEF official start date. The fraction of block trades dropped dramatically after CFTC finalized minimum block size definitions on July 30, 2013 (Fig. 2, Panel F). Thus, we construct *Blkszdt*, a dummy variable that equals one on or after July 30, 2013 and equals zero otherwise. We use the interaction between *Block* and *Blkszdt* to estimate block trade effects occurring after July 30.

In the transaction-level regressions, we continue to control for variations related to segments, product charac-

teristics, and trade information known at trade execution: index dummies, *OTR*, *Tenor5*, *Mty*, *Coupon*, $\text{Log}(\text{TrdSz})$, and *Intrvl*. We calculate statistical significance using heteroskedasticity-consistent standard errors clustered at the index level. The regression results are reported in Table 8. We find strong evidence that centrally cleared trades, SEF trades, end-user trades, and bespoke trades are associated with lower relative effective spread at the transaction level. The coefficients on *Clear*, *Venue*, *Eu*, and *Bespoke* are consistently negative and statistically significant across all specifications in both spread-quoted and price-quoted samples. In the spread-quoted sample, the relative effective spread is lower by 0.45% for cleared trades compared to noncleared trades. Market participants that execute trades on SEFs incur relative effective spreads that are lower by 0.47%. Reductions in relative effective spread for end-user trades and bespoke trades are 0.69% and 0.81%, respectively (Table 8, column 1). In the price-quoted sample, trading costs are lower for cleared, SEF, end-user, and bespoke trades by 0.11%, 0.13%, 0.08%, and 0.16%, respectively (Table 8, column 3).

The coefficient on $\text{Venue} \times \text{Sefdt}$ is negative and significant in the price-quoted sample, implying that the trading cost reduction associated with SEF trading becomes more pronounced after their official start date (Table 8, column 4). The coefficient on $\text{Block} \times \text{Blkszdt}$ is statistically insignificant in both spread-quoted and price-quoted samples, implying that the CFTC's adoption of minimum block sizes did not affect the relation between trading costs and block size.

CDS market segments, trading activity, and product characteristics continue to play a role in explaining trading cost variations at the transaction level. *OTR* and *Tenor5* are negative and statistically significant in both spread- and price-quoted samples. Thus, trades in on-the-run (5-year) indexes incur lower transaction costs than trades in off-the-run (non-5-year) indexes. The coefficient on $\text{Log}(\text{TrdSz})$ is negative and significant, consistent with prior studies of trading costs in OTC fixed income markets. Product characteristics as captured by *Mty* and *Coupon* register as statistically significant explanatory variables in all specifications.

6.2.3. Regression by index family

In the preceding section, we show significant relations between the Dodd-Frank deal characteristics and relative effective spread by pooling information from trades executed in all indexes. We now estimate the same transaction-level regression for four specific index families: IG, HY, Eur, and Xover. Our focus on these families is motivated by a couple of reasons. First, these four indexes represent investment-grade and high-yield issuers in North America and Europe, which are arguably some of the most important corporate credit segments in the global fixed income market. Second, contracts in these index families are subject to the Dodd-Frank Act's mandatory clearing regime. Because we are estimating regressions at the index level, we use heteroskedasticity-robust standard errors and exclude index dummies. The results are reported in Table 9.

Our key findings continue to hold for each of the four major indexes. Liquidity in the four indexes is related to

³¹ The combination of *Eu* and *Misseu* allows us to use all trades in the regressions even if end-user status data are unavailable for some trades.

Table 8

Transaction cost regressions.

This table reports regressions of transaction-level relative effective spread on the Dodd-Frank determinants while controlling for the effects of segments, trading activity, and product characteristics. The dependent variable is transaction-level relative effective spread. The regression sample consists of 158,735 index CDS trades executed between December 31, 2012 and December 31, 2013, publicly disseminated by DDR, and for which the relative effective spread can be calculated. We estimate regressions separately for transactions in spread-quoted and price-quoted contracts. There are 111,491 (47,244) transactions in spread-quoted (price-quoted) contracts. Explanatory variables are constructed from DDR trade reports and capture various Dodd-Frank reforms, and the effects of segments, trading activity, and product characteristics. *Clear* equals one if the trade is centrally cleared and zero otherwise. *Venue* equals one if the trade is executed on a swap execution facility or designated contract market and zero otherwise. *Sefdt* equals one if the trade date occurs on or after October 2, 2013, and zero otherwise. *Venue* × *Sefdt* is the interaction of *Venue* and *Sefdt*. *Eu* equals one for an end-user trade and zero otherwise. If the indication of end-user status is missing, we set *Eu* to zero, and we set a missing end-user status dummy variable, *Misseu*, to one. Trades with end-user status data have *Misseu* set to zero. *Bespoke* equals one for a bespoke trade and zero otherwise. *Block* equals one if the trade is reported as a block trade and zero otherwise. *Blkszdt* equals one on or after July 30, 2013 and zero otherwise. *Block* × *Blkszdt* is the interaction of *Block* and *Blkszdt*. *OTR* equals one if the index series is the most recent series, and equals zero otherwise. *Tenor5* equals one if the index tenor is 5 years, and equals zero otherwise. $\text{Log}(\text{TrdSz})$ is natural log of trade size. *Intrvl* is the time elapsed between consecutive trades (in days). *Mty* is the time between index expiration day and trade execution day (in years). *Coupon* is index series coupon (in basis points). The column labeled Spread (Price) reports regression results based on spread-quoted (price-quoted) contracts. All regressions include index dummies whose coefficients are suppressed for brevity. Coeff is the regression coefficient and SE is the heteroskedasticity-consistent standard error clustered at the index level. *N* is the number of observations. *, **, *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Spread (1)		Spread (2)		Price (3)		Price (4)	
	Coeff	SE	Coeff	SE	Coeff	SE	Coeff	SE
Dodd-Frank determinants								
<i>Clear</i>	−0.445***	(0.120)	−0.470***	(0.115)	−0.106***	(0.003)	−0.103***	(0.004)
<i>Venue</i>	−0.473**	(0.162)	−0.397***	(0.088)	−0.126***	(0.010)	−0.023**	(0.005)
<i>Venue</i> × <i>Sefdt</i>			−0.287	(0.282)			−0.182***	(0.006)
<i>Sefdt</i>			0.302**	(0.109)			0.115***	(0.008)
<i>Eu</i>	−0.686***	(0.096)	−0.693***	(0.094)	−0.076***	(0.002)	−0.063***	(0.001)
<i>Misseu</i>	0.288**	(0.103)	0.325**	(0.124)	0.046*	(0.016)	0.058**	(0.017)
<i>Bespoke</i>	−0.809***	(0.193)	−0.782***	(0.179)	−0.162**	(0.037)	−0.161***	(0.027)
<i>Block</i>	−0.237	(0.148)	0.284	(0.205)	−0.077***	(0.005)	−0.135	(0.180)
<i>Block</i> × <i>Blkszdt</i>			0.164	(0.205)			0.009	(0.197)
<i>Blkszdt</i>			0.486***	(0.149)			−0.110	(0.182)
Controls								
<i>OTR</i>	−1.946***	(0.382)	−1.862***	(0.401)	−0.202***	(0.021)	−0.182***	(0.023)
<i>Tenor5</i>	−1.162***	(0.162)	−1.179***	(0.181)	−0.123***	(0.021)	−0.070*	(0.022)
$\text{Log}(\text{TrdSz})$	−0.247***	(0.056)	−0.285***	(0.053)	−0.021**	(0.004)	−0.020**	(0.004)
<i>Intrvl</i>	0.008	(0.010)	0.007	(0.009)	−0.002	(0.001)	−0.003*	(0.001)
<i>Mty</i>	−0.481***	(0.074)	−0.506***	(0.064)	0.053***	(0.005)	0.033***	(0.005)
<i>Coupon</i>	0.012***	(0.004)	0.011**	(0.004)	−0.001***	(0.000)	−0.002***	(0.000)
<i>Constant</i>	6.590***	(0.342)	6.147***	(0.461)	1.199***	(0.118)	1.339***	(0.030)
Index dummies	Yes		Yes		Yes		Yes	
<i>N</i>	111,491		111,491		47,244		47,244	
Adjusted <i>R</i> ²	0.0668		0.0693		0.0524		0.0573	

the Dodd-Frank deal characteristics. Most of our key variables, such as *Clear*, *Venue*, *Eu*, and *Bespoke*, have negative and statistically significant coefficients, suggesting that trading costs in these indexes are lower for cleared trades, trades executed on SEFs, end-user trades, and bespoke trades. For example, the coefficient on *Clear* ranges from −0.10 (HY) to −0.61 (IG).³² In addition, for the IG, HY, and Eur samples, the coefficient on *Venue* × *Sefdt* is negative and statistically significant. This suggests that market participants trading these indexes benefited from the pick-up in SEF trading after the official start date of October 2, 2013. Results for the control variables indicate that segments, trading activity, and product characteristics help explain trading cost variations within indexes.

The index level regressions show that the liquidity effects associated with the Dodd-Frank determinants are present not only in the entire sample, but also in the most important indexes. In addition, by focusing on these key indexes, we can conduct further tests of clearing-related Dodd-Frank determinants, a topic that we turn to in the next section.

6.3. Phased introduction of contracts subject to mandatory clearing

The CFTC implemented mandatory central clearing in three phases for IG, HY, Eur, and Xover (see Section 4.2 and Panels A and B of Fig. 2). Each phase covers a different category of market participants. Due to its mandatory nature, the phased introduction of central clearing generates exogenous shifts in the economic forces that link clearing, end-user status, and bespoke contracts to transaction costs. Therefore, we use the phased introduction as a natural experiment to test the clearing, end-user, and bespoke hypotheses (Hypotheses 2.1, 2.3, and 2.4), thereby

³² The regression coefficients for CDX NA HY are typically smaller in absolute magnitude, consistent with the fact that the HY index is a price-quoted index. The coefficients on *Tenor5* and *Coupon* cannot be estimated for iTraxx Europe Crossover because all trades in this index are in the 5-year tenor with a coupon of 500 basis points.

Table 9

Regressions by index.

This table reports regressions of transaction-level relative effective spread on the Dodd-Frank determinants and control variables for the following indexes: CDX NA IG (IG), CDX NA HY (HY), iTraxx Europe (Eur), and iTraxx Europe Crossover (Xover). The dependent variable is transaction-level relative effective spread. The regressions employ 46,422, 41,868, 29,024, and 21,764 trades, respectively, in IG, HY, Eur, and Xover executed between December 31, 2012 and December 31, 2013, publicly disseminated by DDR, and for which the relative effective spread can be calculated. Explanatory variables are constructed from DDR trade reports and capture various Dodd-Frank reforms, and the effects of segments, trading activity, and product characteristics. *Clear* equals one if the trade is centrally cleared and zero otherwise. *Venue* equals one if the trade is executed on a swap execution facility or designated contract market and zero otherwise. *Sefdt* equals one if the trade date occurs on or after October 2, 2013, and zero otherwise. *Venue* $\times$ *Sefdt* is the interaction of *Venue* and *Sefdt*. *Eu* equals one for an end-user trade and zero otherwise. If the indication of end-user status is missing, we set *Eu* to zero, and we set a missing end-user status dummy variable, *Misseu*, to one. Trades with end-user status data have *Misseu* set to zero. *Bespoke* equals one for a bespoke trade and zero otherwise. *Block* equals one if the trade is reported as a block trade and zero otherwise. *Blkszdt* equals one on or after July 30, 2013 and zero otherwise. *Block* $\times$ *Blkszdt* is the interaction of *Block* and *Blkszdt*. *OTR* equals one if the index series is the most recent series, and equals zero otherwise. *Tenor5* equals one if the index tenor is 5 years, and equals zero otherwise. $\text{Log}(\text{TrdSz})$ is natural log of trade size. *Intrvl* is the time elapsed between consecutive trades (in days). *Mty* is the time between index expiration day and trade execution day (in years). *Coupon* is index series coupon (in basis points). Coeff is the regression coefficient and SE is the heteroskedasticity-consistent standard error. *N* is the number of observations. *, **, *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	IG		HY		Eur		Xover	
	Coeff	SE	Coeff	SE	Coeff	SE	Coeff	SE
Dodd-Frank determinants								
<i>Clear</i>	−0.605***	(0.045)	−0.100***	(0.006)	−0.483***	(0.057)	−0.085	(0.066)
<i>Venue</i>	−0.303**	(0.132)	−0.027***	(0.009)	−0.193**	(0.089)	−0.577***	(0.223)
<i>Venue</i> $\times$ <i>Sefdt</i>	−0.865***	(0.149)	−0.184***	(0.013)	−0.206*	(0.120)	0.037	(0.231)
<i>Sefdt</i>	0.444***	(0.079)	0.122***	(0.010)	0.561***	(0.091)	0.627***	(0.116)
<i>Eu</i>	−0.691***	(0.059)	−0.065***	(0.008)	−0.482***	(0.077)	−0.970***	(0.090)
<i>Misseu</i>	0.509***	(0.059)	0.045***	(0.008)	0.503***	(0.071)	−0.027	(0.091)
<i>Bespoke</i>	−0.640***	(0.073)	−0.126***	(0.011)	−0.252**	(0.111)	−0.981***	(0.097)
<i>Block</i>	−0.050	(0.210)	−0.352***	(0.076)	0.352***	(0.124)	0.328*	(0.193)
<i>Block</i> $\times$ <i>Blkszdt</i>	0.501**	(0.227)	0.239***	(0.077)	−0.124	(0.173)	0.280	(0.220)
<i>Blkszdt</i>	0.487**	(0.215)	−0.331***	(0.076)	0.354***	(0.137)	0.094	(0.203)
Controls								
<i>OTR</i>	−1.066***	(0.104)	−0.165***	(0.013)	−2.665***	(0.213)	−2.475***	(0.229)
<i>Tenor5</i>	−3.374***	(0.503)	−0.084	(0.069)	−2.822***	(0.766)		
$\text{Log}(\text{TrdSz})$	−0.211***	(0.019)	−0.017***	(0.002)	−0.219***	(0.029)	−0.507***	(0.040)
<i>Intrvl</i>	0.019	(0.058)	−0.002**	(0.001)	−0.001	(0.014)	−0.016	(0.189)
<i>Maturity</i>	−0.902***	(0.101)	0.035***	(0.008)	−0.442***	(0.132)	−1.324***	(0.349)
<i>Coupon</i>	0.109***	(0.026)	−0.001	(0.001)	−0.025*	(0.013)		
Constant	0.308	(1.689)	1.433***	(0.530)	11.942***	(2.468)	11.722***	(1.701)
<i>N</i>	46,422		41,868		29,024		21,764	
Adjusted <i>R</i> ²	0.0796		0.0522		0.0658		0.0715	

avoiding contamination by endogeneity effects. We implement the phased introduction test by estimating the following transaction-level RES regression for each of the four indexes

$$\begin{aligned}
 \text{RES} = & f(\text{Clear} \times \text{Pre}, \text{Clear} \times \text{Phase1}, \text{Clear} \times \text{Phase2}, \\
 & \text{Clear} \times \text{Phase3}, \text{Eu} \times \text{Pre}, \\
 & \text{Eu} \times \text{Phase1}, \text{Eu} \times \text{Phase2}, \text{Eu} \times \text{Phase3}, \\
 & \text{Bespoke} \times \text{Pre}, \text{Bespoke} \times \text{Phase1}, \text{Bespoke} \times \text{Phase2}, \\
 & \text{Bespoke} \times \text{Phase3}, \\
 & \text{Misseu}, \text{Venue}, \text{Venue} \times \text{Sefdt}, \\
 & \text{Sefdt}, \text{Block}, \text{Block} \times \text{Blkszdt}, \text{Blkszdt}, \text{Phase1}, \\
 & \text{Phase2}, \text{Phase3}, \text{OTR}, \text{Tenor5}, \text{Log}(\text{TrdSz}), \\
 & \text{Intrvl}, \text{Mty}, \text{Coupon}),
 \end{aligned} \quad (5)$$

where each observation is for a transaction in an index CDS contract (defined by index, series, and tenor) during the sample period. The specification allows for different liquidity effects across successive phase-in periods by using dummy variables. *Pre* equals one before the start of Category 1 mandatory clearing, and equals zero otherwise. *Phase1* equals one when Category 1 mandatory clearing is in effect but before the start of Category 2 manda-

tory clearing, and equals zero otherwise. *Phase2* equals one when Category 2 mandatory clearing is in effect but before the start of Category 3 mandatory clearing, and equals zero otherwise. *Phase3* equals one when Category 3 mandatory clearing is in effect, and equals zero otherwise. [Appendix A.2.1](#) describes market participants in each of Categories 1, 2, and 3 and provides mandatory clearing start dates for each category. We interact the phase dummies with three Dodd-Frank deal characteristics: *Clear*, *Eu*, and *Bespoke*. Each interaction captures the incremental effect of a given characteristic on trading costs during a particular implementation phase. For example, *Eu* $\times$ *Phase1* captures the difference in relative effective spreads between an end-user trade and a non-end-user trade during Phase 1. These interaction terms form the focus of the regression analyses. [Table 10](#) reports the regression results.

Evidence from the phased introduction of mandatory clearing is consistent with the hypotheses that centrally cleared trades, end-user trades, and bespoke trades have lower transaction costs. The liquidity improvements associated with these deal characteristics are evident across all four indexes after mandatory clearing was fully phased-in. The coefficients on *Clear* $\times$ *Phase2* and *Clear* $\times$ *Phase3* are negative and statistically significant, indicating that

Table 10

Mandatory clearing phase-in analysis.

This table reports regressions of transaction-level relative effective spread on the Dodd-Frank determinants and control variables for the following indexes: CDX NA IG (IG), CDX NA HY (HY), iTraxx Europe (Eur), and iTraxx Europe Crossover (Xover). The dependent variable is transaction-level relative effective spread. The regressions employ 46,422, 41,868, 29,024, and 21,764 trades, respectively, in IG, HY, Eur, and Xover executed between December 31, 2012 and December 31, 2013, publicly disseminated by DDR, and for which the relative effective spread can be calculated. Explanatory variables are constructed from DDR trade reports and capture various Dodd-Frank reforms, and the effects of segments, trading activity, and product characteristics. *Clear* equals one if the trade is centrally cleared and zero otherwise. *Venue* equals one if the trade is executed on a swap execution facility or designated contract market and zero otherwise. *Sefdt* equals one if the trade date occurs on or after October 2, 2013, and zero otherwise. *Venue* $\times$ *Sefdt* is the interaction of *Venue* and *Sefdt*. *Eu* equals one for an end-user trade and zero otherwise. If the indication of end-user status is missing, we set *Eu* to zero, and we set a missing end-user status dummy variable, *Misseu*, to one. Trades with end-user status data have *Misseu* set to zero. *Bespoke* equals one for a bespoke trade and zero otherwise. *Block* equals one if the trade is reported as a block trade and zero otherwise. *Blkszdt* equals one on or after July 30, 2013 and zero otherwise. *Block* $\times$ *Blkszdt* is the interaction of *Block* and *Blkszdt*. *Pre* equals one before the start of Category 1 mandatory clearing and zero otherwise. *Phase1* equals one when Category 1 mandatory clearing is in effect but before the start of Category 2 mandatory clearing and zero otherwise. *Phase2* equals one when Category 2 mandatory clearing is in effect but before the start of Category 3 mandatory clearing and zero otherwise. *Phase3* equals one when Category 3 mandatory clearing is in effect and zero otherwise. *Pre*, *Phase1*, *Phase2*, and *Phase3* adjust for the fact that CDX indexes (IG and HY) and iTraxx indexes (Eur and Xover) have different mandatory clearing start dates. *OTR* equals one if the index series is the most recent series, and equals zero otherwise. *Tenor5* equals one if the index tenor is 5 years, and equals zero otherwise. Trading activity variables are *Log(TrdSz)* and *Intrvl*. *Log(TrdSz)* is the natural log of trade size. *Intrvl* is the time elapsed between consecutive trades (in days). Product characteristics are *Mty* and *Coupon*. *Mty* is the time between index expiration day and trade execution day (in years). *Coupon* is index series coupon (in basis points). For brevity, we report regression coefficients for the interactions between the phase-in dummies and *Clear*, *Eu*, and *Bespoke*. All other regression coefficients are suppressed. Coeff is the regression coefficient and SE is the heteroskedasticity-consistent standard error. *N* is the number of observations. *, **, *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	IG		HY		Eur		Xover	
	Coeff	SE	Coeff	SE	Coeff	SE	Coeff	SE
Dodd-Frank determinants								
<i>Clear</i> $\times$ <i>Pre</i>	−0.280***	(0.054)	−0.122***	(0.009)	−0.517***	(0.055)	−0.430***	(0.067)
<i>Clear</i> $\times$ <i>Phase1</i>	0.626***	(0.096)	0.069***	(0.012)	0.279**	(0.130)	1.145***	(0.144)
<i>Clear</i> $\times$ <i>Phase2</i>	−1.320***	(0.097)	−0.066***	(0.011)	−1.865***	(0.127)	−1.120***	(0.134)
<i>Clear</i> $\times$ <i>Phase3</i>	−2.316***	(0.094)	−0.391***	(0.015)	−0.940***	(0.137)	−1.602***	(0.139)
<i>Eu</i> $\times$ <i>Pre</i>	0.304***	(0.087)	0.035***	(0.013)	−0.044	(0.094)	−0.122	(0.138)
<i>Eu</i> $\times$ <i>Phase1</i>	−0.303***	(0.094)	0.029**	(0.014)	−0.109	(0.137)	−0.549***	(0.138)
<i>Eu</i> $\times$ <i>Phase2</i>	−1.681***	(0.094)	−0.085***	(0.013)	−1.809***	(0.147)	−1.783***	(0.161)
<i>Eu</i> $\times$ <i>Phase3</i>	−1.278***	(0.087)	−0.170***	(0.013)	−0.708***	(0.143)	−1.352***	(0.132)
<i>Bespoke</i> $\times$ <i>Pre</i>	0.014	(0.223)	−0.073***	(0.027)	1.030***	(0.282)	0.029	(0.231)
<i>Bespoke</i> $\times$ <i>Phase1</i>	−0.287*	(0.154)	−0.059***	(0.017)	0.209	(0.214)	−0.858***	(0.130)
<i>Bespoke</i> $\times$ <i>Phase2</i>	−1.257***	(0.114)	−0.073***	(0.020)	−2.110***	(0.159)	−1.629***	(0.200)
<i>Bespoke</i> $\times$ <i>Phase3</i>	−1.573***	(0.146)	−0.368***	(0.018)	−0.873***	(0.205)	−1.866***	(0.154)
<i>Misseu</i>	0.409***	(0.059)	0.045***	(0.008)	0.363***	(0.074)	0.075	(0.098)
<i>Venue</i> , <i>Venue</i> $\times$ <i>Sefdt</i> , <i>Sefdt</i>	Yes		Yes		Yes		Yes	
<i>Block</i> , <i>Block</i> $\times$ <i>Blkszdt</i> , <i>Blkszdt</i>	Yes		Yes		Yes		Yes	
<i>Phase1</i> , <i>Phase2</i> , <i>Phase3</i>	Yes		Yes		Yes		Yes	
Controls								
<i>OTR</i> and <i>Tenor5</i>	Yes		Yes		Yes		Yes	
Trading activity variables	Yes		Yes		Yes		Yes	
Product characteristics	Yes		Yes		Yes		Yes	
Constant	Yes		Yes		Yes		Yes	
<i>N</i>	46,422		41,868		29,024		21,764	
Adjusted <i>R</i> ²	0.1087		0.0805		0.0866		0.0984	

cleared trades have lower relative effective spreads than noncleared trades when mandatory clearing was extended to Categories 2 and 3 market participants, respectively. For example, the average cleared trade in IG incurs a relative effective spread that is lower by 1.32% (2.32%) during Phase 2 (3). Trading costs are also lower for end-user and bespoke trades when mandatory clearing was fully phased-in. For example, the relative effective spread declines by 1.68% and 1.28% (1.26% and 1.57%), respectively, for the average end-user trade (bespoke trade) in IG during Phases 2 and 3. There is some evidence indicating trading cost reductions for end-users and users of bespoke contracts during Phase 1.

Using the phased introduction of mandatory clearing as a natural experiment, we continue to find beneficial liquidity effects for cleared, end-user, and bespoke trades. These

results, together with those from the daily and transaction-level regressions, provide strong support for our hypotheses.

7. Robustness checks

We conduct a series of robustness checks to confirm the robustness of our results. (For brevity, the results of these robustness checks are not presented in the paper, but are available upon request.) First, while the first year of the Dodd-Frank regulatory regime provides the most critical time period for evaluating the impact of regulatory reforms, it is also interesting to see if the relations between various Dodd-Frank deal characteristics and index CDS transaction costs and liquidity persist in an extended period after the first year. Therefore, we collect

additional data and extend our sample to May 31, 2014. Using the extended data set, we repeat the analyses reported in Tables 7–9 and find that our reported results (based on 2013 data) are confirmed in a period that goes beyond the first year of the Dodd-Frank regulatory regime. Second, similar to the phased introduction test of Section 6.3 (see Table 10), for each of the four major indexes we conduct an event study of the four key daily liquidity measures (i.e., *RES*, *Log(Amihud)*, *Disp*, and *Roll*) around the three phases of the introduction of mandatory clearing by the CFTC. The results are consistent with the phased introduction test results in Table 10 and confirm that the effects of Dodd-Frank deal characteristics on key liquidity measures do not arise due to differences in the nature of the contracts themselves or to endogeneity concerns. Third, Pelizzon, Subrahmanyam, Tomio, and Uno (2015) use threshold analysis to study the nonlinearity of the relation between credit risk and market liquidity in the sovereign credit market and conduct structural break tests to confirm that central bank monetary policy interventions affect the credit risk-liquidity relation. In keeping with Pelizzon, Subrahmanyam, Tomio, and Uno (2015), we conduct structural break tests for the daily regression specification of Section 6.2.1 to check if the introduction dates of the new phases of mandatory clearing represent points of discontinuity. To ensure that the results are not driven by differences in the nature of the index CDS contracts, we conduct the structural break tests separately for each of the four major indexes. We find strong evidence that the introduction dates represent points of discontinuity in the relations between the key liquidity measures and explanatory variables including Dodd-Frank deal characteristics.

8. Conclusion

In this paper, we provide a first look at trading costs and liquidity in the OTC derivatives market using the newly available real-time trade reports for index CDSs, an important class of OTC derivatives. Using a merged data set of intraday trades and quotes, we find that the average relative effective spread is about 0.27% of price level or 2.73% of CDS spread. In addition, we explore how trading costs are related to Dodd-Frank regulatory reforms. After the commencement of real-time reporting and public dissemination of OTC derivatives trades, we find strong evidence of liquidity improvement in the index CDS market using a wide variety of liquidity measures including bid-ask spread, price impact, and price dispersion. In addition, we find that trade characteristics linked to Dodd-Frank regulatory reforms are associated with lower trading costs and hence higher liquidity. Specifically, trading costs are lower for centrally cleared trades and trades executed on SEFs. Customized trades and trades executed by end-users for hedging purposes also have lower trading costs. We obtain similar results using alternative liquidity measures such as price impact and price dispersion. We also find liquidity variations across different segments in the index CDS market. The most actively traded indexes, on-the-run index series, and 5-year index contracts have higher liquidity and lower trading costs. Our study provides a better under-

standing of the multi-trillion dollar OTC derivatives market and should be of interest to a diverse audience.

Appendix A. Real-time trade reporting requirements for OTC swaps and DDR data fields associated with Dodd-Frank regulatory reforms

This appendix provides a brief summary of the Dodd-Frank real-time trade reporting requirements and the DDR data fields that are associated with various Dodd-Frank regulatory reforms.

A.1. Dodd-Frank real-time trade reporting requirement and its implementation by the CFTC

The Dodd-Frank Act requires swap counterparties to report transactions in real-time to SDRs, which in turn disseminate the transaction data to the public. An SDR is “any person that collects and maintains information or records with respect to transactions or positions in, or the terms and conditions of, swaps entered into by third parties for the purpose of providing a centralized recordkeeping facility for swaps” (Dodd-Frank Act Section 721(a)(48)).

Section 727 of the Dodd-Frank Act charges the CFTC with the responsibility of implementing mandatory real-time public reporting of transaction data for index CDSs (and other credit derivatives), interest rate swaps, equity swaps, foreign exchange swaps, and other commodity swaps. The CFTC defines real-time public reporting as “the reporting of data relating to a swap transaction, including price and volume, as soon as technologically practicable after the time at which the swap transaction has been executed” (CFTC Regulation Section 43.2).

On December 31, 2012, registered swap dealers commenced reporting transaction data for index CDSs and interest rate swaps. On February 28, 2013, they also started reporting transaction data for the remaining classes of swaps: equity, foreign exchange, and other commodity swaps. Finally, on April 10, 2013, all other swap market participants had to comply with real-time reporting for all five classes of swaps. Background information on the determination of these compliance dates can be found in CFTC Regulation Part 43, CFTC No Action Letters No. 12–32 (November 19, 2012) and No.12–41 (December 5, 2012). Market participants report their swap transactions to SDRs, which in turn disseminate the data to the public. For each transaction, only one of the counterparties is responsible for reporting. Thus, the issue of double counting of trades does not arise (CFTC Regulation Section 43.3).

A.2. DDR data fields associated with the Dodd-Frank regulatory reforms

The index CDS trade data used in the paper are obtained from DDR, which publicly disseminates OTC trade reports in accordance with the regulations described above. In addition to transaction details such as trade price, trade size, and timestamp, each DDR trade report provides information on a range of characteristics linked to the regulatory reforms. In the following discussion, we identify

the various aspects of the Dodd-Frank regulatory reforms and their associated data fields in the DDR trade reports.

A.2.1. Central clearing

The CFTC requires reporting counterparties to indicate whether the trade is cleared by a central counterparty in the data field, “CLEAR.” The CFTC implemented mandatory clearing in 2013 for a set of index CDSs: CDX NA IG, CDX NA HY, iTraxx Europe, iTraxx Europe Crossover, and iTraxx Europe HiVol. For the CDX indexes, the start date of mandatory clearing is March 11, June 10, and September 9, respectively, for Category 1 entities (e.g., swap dealers), Category 2 entities (e.g., commodity pools), and Category 3 entities (all other swap market participants). For the iTraxx indexes, the start dates for Categories 1, 2, and 3 are, respectively, April 26, July 25, and October 23. According to CFTC Regulation Parts 39 and 50, the following series and tenors are covered by the clearing requirements:

- CDX.NA.IG 3Y: Series 15 and all subsequent series.
- CDX.NA.IG 5Y: Series 11 and all subsequent series.
- CDX.NA.IG 7Y: Series 8 and all subsequent series.
- CDX.NA.IG 10Y: Series 8 and all subsequent series.
- CDX.NA.HY 5Y: Series 11 and all subsequent series.
- iTraxx Europe 5Y: Series 10 and all subsequent series.
- iTraxx Europe 10Y: Series 7 and all subsequent series.
- iTraxx Europe Crossover 5Y: Series 10 and all subsequent series.
- iTraxx Europe HiVol 5Y: Series 10 and all subsequent series.

A.2.2. SEF

Reporting counterparties must indicate whether the trade is executed at an SEF in the data field, “EXECUTION_VENUE.” According to Section 721 of the Dodd-Frank Act, an SEF is a trading venue in which multiple traders can execute swap transactions by accepting bids and offers posted by other traders. Thus, an SEF is intended to provide a multiple-to-multiple trading environment where customers can trade not just with dealers, but potentially with other customers as well.

A.2.3. End-user

The CFTC requires reporting counterparties to specify whether a party to a swap contract qualifies for end-user status in the data field, “INDICATION_OF_END_USER_EXCEPTION.” In other words, this data field allows researchers to identify trades involving at least one end-user. To qualify as an end-user, a market participant must satisfy a number of conditions. First, the market participant must be a nonfinancial entity that enters into swap transactions to hedge risks associated with its underlying business (e.g., a manufacturer that uses CDSs to hedge the default risk of its key customers). Second, the market participant must explain how it meets financial obligations associated with entering into non-cleared swaps to the CFTC. Third, if the market participant is a publicly traded firm that reports to the Securities and Exchange Commission (SEC), its decision to use noncleared swaps must be reviewed and approved by an appropriate committee of the board of directors. Fourth, the market

participant must provide information about its end-user status to the SDR (Commodity Exchange Act Section 2(h)(7) and CFTC Regulation Section 39.6). An end-user can elect to be exempted from mandatory clearing and SEF trading.

A.2.4. Bespoke (non-standardized) contract

Reporting counterparties have to indicate whether the trade is bespoke (non-standardized) in the data field, “INDICATION_OF_OTHER_PRICE_AFFECTING_TERM.” According to CFTC Regulation Part 43, a bespoke or non-standardized swap is a swap that has one or more additional terms or provisions other than those listed in the real-time report's data fields, and these additional terms materially affect the swap's price. Bespoke contracts are not subject to mandatory clearing and SEF trading.

A.2.5. Block trades

The CFTC requires reporting counterparties to indicate whether the trade is a block trade in the data field, “BLOCK_TRADES_AND_LARGE_NOTIONAL_OFF-FACILITY_SWAPS.” The CFTC's minimum block size definitions for swaps became effective on July 30, 2013. The block size definitions are available at <http://www.cftc.gov/ucm/groups/public/@lrfederalregister/documents/file/2013-12133a.pdf>. The CFTC stipulates different delays for block trades depending on criteria such as counterparty identity (e.g., whether at least one counterparty is a swap dealer), mandatory clearing requirement, and type of swap (CFTC Regulation Section 43.5).

Appendix B. Description of index CDS products

This appendix provides a brief summary of various index CDS contracts and explains how they are constituted.

An index CDS contract is a credit derivatives contract that offers default protection for a collection of reference entities (the index). A variety of indexes, each distinguished by a different set of constituents, form the underlying for index CDS contracts. Different indexes are designed to represent different segments of the global credit market. In this paper, we focus on indexes that represent corporate bond issuers from various geographical regions, US municipal bond issuers, and emerging market sovereign bond issuers. These indexes, which represent the most liquid part of their respective segments, are owned and managed by Markit.³³ The following table describes the bond issuers underlying each index featured in our empirical analysis. Indexes appear in the same order as in Table 3, Panel A.

Index	Underlying issuers
Corporate	
CDX NA IG	125 North American investment grade issuers
CDX NA HY	100 North American high yield issuers

³³ Additional details on index CDSs can be found in Markit (2014), along with the rules for specific indexes. These documents are available to the public at www.markit.com.

Index	Underlying issuers
iTraxx Europe	125 European investment grade issuers
iTraxx Eur	50 most liquid sub-investment grade issuers
Crossover	
iTraxx Eur Sr	25 financial entities from iTraxx Europe referencing senior debt
Finls	
iTraxx Japan	50 Japanese investment grade issuers
iTraxx Eur	25 financial entities from iTraxx Europe referencing subordinate debt
Sub Finls	
iTraxx Asia	40 non-Japanese Asian investment grade issuers
ex Japan IG	
iTraxx	25 investment grade Australian issuers
Australia	
CDX LatAm	20 Latin American issuers
Corp	
iTraxx	25 corporate and quasi-sovereign issuers from Central and Eastern Europe, Middle East, and Africa
CEEMEA	
iTraxx Eur	30 nonfinancial issuers with the widest 5-year CDS spreads from iTraxx Europe
HiVol	
Municipal	
MCDX	50 US municipal issuers
Sovereign	
CDX EM	14 sovereign issuers from Latin America, Eastern Europe, the Middle East, Africa, and Asia, although number of issuers can vary.
	Sovereign issuers from CDX EM that are not part of the European Union
CDX EM	
ex-EU	

All the above indexes are equally weighted, except for CDX EM and CDX EM ex-EU whose weights are determined through dealer polls administered by Markit. Every six months, Markit revises the constituents of each index based on input from dealers. The updated constituents form a new series of the index. A new series starts trading on March 20 or September 20 (roll date), except for MCDX, whose roll date is either April 3 or October 3.³⁴ For example, CDX NA IG Series 19 has a roll date of September 20, 2012, while CDX NA IG Series 20 has a roll date of March 20, 2013.

Appendix C. Matching and cleaning DDR trade reports

In this appendix, we first describe the data collection procedures for CMA intraday quote data and Markit end-of-day valuation data, and then explain how we match DDR trade reports to intraday quotes and end-of-day valuations. We also point out various data issues in the DDR reports and discuss how we address them.

C.1. CMA intraday quotes (tick-by-tick)

CMA obtains intraday CDS dealer quotes (both executable and indicative) by accessing communication between CDS dealers and approximately 40 buy-side institutions, including hedge funds and investment banks' proprietary trading desks. The data collection process for a given contract (defined by index, series, tenor, etc.) begins with each buy-side contributor, from whom CMA obtains and aggregates the most recent quotes received by that contributor from at least a fixed number of distinct dealers. The quote aggregation process also employs measures to

mitigate the influence of outliers. The contributor-level aggregated data are then further aggregated across contributors to derive the tick-by-tick bid quote and offer quote for that contract that are disseminated to contributors in real-time.

C.2. Markit end-of-day valuation

Markit sources daily pricing data from a group of 22 global banks to produce its CDS composite spread, price, and depth for a given contract. The data include daily closing prices in the books of record of contributing banks and/or the most recent prices from these banks' trading desks. When at least two contributed CDS spreads pass its data cleaning tests, Markit averages these numbers to produce its daily CDS composite spread and price. Markit's daily composite depth is the number of contributors who provide data for the calculation of the 5-year CDS composite spread and price.

C.3. Linking DDR to CMA and Markit

The first step of the matching process is to extract RED code and maturity from the DDR report. Created by Markit, the RED code is a 9-character alphanumeric code that uniquely identifies the series and version of an index CDS, and is a widely used security identifier in the CDS market.³⁵ For example, CDX NA IG Series 19 version 1 has RED code "2I65BYCV5." Because index CDS contracts are traded in different maturities (also known as tenors), we combine the RED code with the maturity to uniquely identify the index referenced in the trade report. There are 203,225 trade reports referencing indexes covered by the Markit and CMA data, and we are able to extract the RED code and maturity for 203,165 of these reports.

Next, we use the combination of RED code and maturity to match trades to the Markit and CMA data. In a small number of cases, non-matches arise because maturity dates are misreported in the DDR reports. Index CDSs have standardized maturity dates; for example, the 5-year contract referencing CDX NA IG Series 19 version 1 matures on December 20, 2017. While Markit and CMA data reflect these standardized dates, some DDR reports mistakenly indicate in the maturity date field the trade date or trade confirmation date. After excluding non-matches, we are able to link 202,532 index CDS trade reports to CMA and Markit with RED code and maturity.³⁶

³⁵ We extract the RED code from the "UNDERLYING_ASSET_1" data field and we extract the maturity from the "END_DATE" data field.

³⁶ We also attempt to match index tranche trades to Markit and CMA. We discover a major shortcoming in the CFTC's reporting requirement for index tranches. Index tranches are contractually identified by RED code, maturity, and attachment and detachment points. However, the vast majority of DDR reports for tranches do not provide attachment and detachment points. (Of the 7,839 tranche trades with RED code and maturity, 6,769 trades do not have information on attachment and detachment points.) This makes it essentially impossible for the general public to identify the correct tranche underlying the reported trade, thus rendering most of the tranche reports unusable. Hence, we do not include index tranches in the ensuing analysis. Our finding indicates that the CFTC should explicitly instruct counterparties to report attachment and detachment points for tranche trades.

³⁴ If a roll date falls on a non-business day, then trading begins on the next business day.

C.4. Correcting data issues in DDR trade reports

We begin by retaining reports with quoting convention, standard currency denomination, and end-of-day valuation from Markit and CMA, which yields 200,421 reports. Then, we check each trade report to ascertain the completeness of price information. We find that 11 reports do not have price notation type, 5,587 reports have missing prices, and 14,645 reports have a reported price of zero. Excluding these reports with incomplete price information as well as an additional 29 reports with non-standard currency denominations brings the sample to 180,149 reports.

C.4.1. Correcting data issues in reported trade price³⁷

Among the trade reports that contain complete price information, we further identify potential problems and address various recording errors.

One data issue stems from the confusion between fixed coupon and trade price. Index CDSs in our sample are traded with fixed coupons with 100 or 500 basis points being the most common. We find instances wherein the index's fixed coupon is erroneously reported instead of the trade price. The incorrectly reported value is either 0.01, 1, 100, 0.05, 5, or 500. Thus, we delete a report if the price takes on any of these exact values.

Another data issue concerns quoting convention. In the index CDS market, there are two quoting conventions: quotes are either in the form of spread or price. By matching our trade reports with CMA, we obtain each index's quoting convention. This enables us to identify potential problems and correct errors in reported spreads/prices.

For spread-quoted indexes (which should be measured in basis points), we find cases in which the CDS spread is mistakenly reported in decimal form or in percentage points. Suppose a trade is executed at a spread of 110 basis points (bps). Instead of 110, the counterparty may report 0.011 (decimal form) or 1.10 (percentage). We adjust for these inconsistencies in the following way. If the reported number is less than 0.1, we multiply it by 10,000; if the reported number is at least 0.1 but less than 10, we multiply it by 100.³⁸

For price-quoted indexes (which should be measured in percentages), we find two types of errors. The first type of error has two dimensions. The counterparty misinterprets the price (quoted as a percent of par) as a value measured in basis points. For example, a price of 94% of par is misinterpreted as 94 basis points. To compound the problem, the counterparty reports the misinterpreted price in decimal form. Continuing with the example, 94 is read as 94 basis points and reported as 0.0094. To correct these cases, we multiply by 10,000 any price that is greater than 0.008 but less than 0.012. The second type of error is that the trade price is reported in decimal form. For example, a price of 94% of par is reported as 0.94. To correct these

cases, we multiply by 100 any price that is greater than 0.8 but less than 1.2.³⁹

C.4.2. Correcting data issues in reported trade size

There is a lack of uniformity in the reporting of a trade's notional amount (i.e., trade size). For US dollar-denominated contracts, we find that trade size is reported in millions of dollars (i.e., 10 million notional is reported as 10) or in dollars (i.e., 10 million notional is reported as 10,000,000). Both approaches seem to be acceptable according to the language of CFTC's real-time reporting regulation. CFTC Regulation Part 43 Table A1 suggests that notional amounts are to be expressed in millions of dollars, while Section 43.4 of the same regulation implies that notional amounts have to be expressed in dollars. The lack of clarity and consistency in the regulation produces reported trade sizes that appear orders of magnitude apart. Given that trade sizes typically run in the millions of dollars in the CDS market, when a reported notional amount does not exceed 100, we multiply it by one million. Our analysis of trade size reporting underscores the need for the CFTC to unambiguously specify the reporting format for the notional amount so as to enhance the usability of the real-time trade reports.

C.4.3. Correcting for anomalous deviations from market consensus valuation

As a final step to remove potentially erroneous trade reports, we filter out extreme prices. For spread-quoted indexes, we keep all reports where the absolute percentage difference between reported price and Markit's end-of-day spread does not exceed 25%. For price-quoted indexes, we keep all reports where the absolute percentage difference between reported price and Markit's end-of-day price does not exceed 2.5%.

After applying the above data cleaning filters, we obtain the final sample of 165,919 trades executed between December 31, 2012 and December 31, 2013, of which 117,377 are trades in spread-quoted contracts, and 48,542 are trades in price-quoted contracts.

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³⁷ Price is reported in the "PRICE_NOTATION" data field; volume is reported in the "NROUNDED_NOTIONAL_AMOUNT_1" data field.

³⁸ The cut-off points are derived from visually inspecting the data and the fact that CDS spreads below 10 bps or above 1,000 bps are extremely unlikely for spread-quoted index CDSs during our sample period.

³⁹ The cut-off points are based on visual inspection of the data and the fact that prices below 80% or above 120% are extremely unlikely for price-quoted index CDSs during our sample period.

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